



Verified Carbon Standard

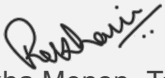
VALIDATION REPORT

ELMADAG BIOGAS POWER PLANT

Document Prepared by



Project Title	Elmadag Biogas Power Plant
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Report Title	Validation report of "Elmadag Biogas Power Plant"
Client	ABRAK ENERJİ ELEKTRİK ÜRETİM ANONİM ŞİRKETİ
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Summary:

ABRAK ENERJİ ELEKTRİK ÜRETİM ANONİM ŞİRKETİ has commissioned EcoLance Pvt. Ltd to carry out the validation of the **“Elmadag Biogas Power Plant” project (hereafter referred to as the Project)**. This report summarizes the findings of the validation of the project, performed on the basis of VCS standard Version 4.5 criteria, GHG program applied as well as criteria given to provide for consistent project operations, monitoring and reporting. For the validation procedure a desk review of the project design documents, follow-up interviews with the local stakeholders, and resolution of outstanding issues methods were followed. EcoLance confirms that the review of the project design, supporting documents and the subsequent follow-up interviews have provided sufficient evidence to determine the fulfilment of stated criteria.

The purpose of the proposed project activity is to implement a methane capture and utilization system that uses the biogas potential of animal manure generated at the neighboring farms. This biogas-to-energy project that will generate renewable energy by capturing biogas from animal manure -via anaerobic digestion- and utilizing it to produce thermal and electric energy through gas engines. The project enables reduction of GHG incurred from existing system of cattle manure generated at farms, which is left to decay at and around farms in anaerobic conditions. **According to the provisional acceptance protocol, the start date of the operation of the project is 31/12/2020 and the installed capacity of the project is 2.406 MWe.**

The objective of the Validation is to have an independent evaluation of a project activity by a VVB against the requirements of the VCS and GHG program applied, on the basis of the project design document. In particular, the project's baseline, monitoring plan, and the project's compliance with relevant VCS requirements, GHG program requirements and host Party criteria are validated in order to confirm that the project design, as documented, is sound and reasonable and meets the identified criteria. Validation is a requirement for all VCS projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of Verified Carbon Units (VCUs). **Validation is not meant to provide any consultancy towards the project participants. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the project design.**

The validation team has raised **2 CAR, and 6 CLs**. While CARs and CLs have been successfully closed, **01 FAR** has been raised. More information is provided in Appendix-2 of the report. In conclusion, it is EcoLance's opinion that the project activity "Elmadag Biogas Power Plant" project, as described in the VCS-PD version 3 of **08/12/2023**, meets all relevant requirements for VCS activities and all relevant Host Party criteria and correctly applies the small-scale methodology "AMS III.D: Methane recovery in animal manure management systems", Version 21.0 and "AMS I.D: Grid connected renewable electricity generation", Version 18.0.

Validation Conclusion:

The conclusions of this report show that the "Elmadag Biogas Power Plant" , as it was described in the VCS-PD, is in line with all **relevant UNFCCC requirements for the CDM, VCS and all relevant host part criteria and correctly applies the small-scale methodology "AMS-III.D: Methane recovery in animal manure management systems", Version 21.0 and "AMS-I.D: Grid connected renewable electricity generation", Version 18.0**. This validation is based on the information made available to EcoLance and the engagement conditions are detailed in this report. No restrictions or uncertainties were identified related to the validation. Hence, EcoLance request the registration of the proposed project activity as a VCS project activity.

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1 INTRODUCTION

1.1 Objective

The objective of the Validation is to have an independent evaluation of a project activity by a VVB against the requirements of the VCS and GHG program applied, on the basis of the VCS-PD version 03 dated 08/12/2023. In particular, the project's baseline, monitoring plan, and the project's compliance against the small-scale methodology "AMS-III.D: Methane recovery in animal manure management systems", Version 21.0 and "AMS-I.D: Grid connected renewable electricity generation", Version 18.0, with requirements of Article 12 of the Kyoto Protocol, the CDM Modalities and Procedures with relevant VCS requirements, GHG program requirements and Host Party criteria are validated in order to confirm that the project design, as documented, is sound and reasonable and meets the identified criteria. Validation is a requirement for all VCS projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of Verified Carbon Units (VCUs).

1.2 Scope and Criteria

The validation scope is given as an independent and objective review of the project design, the project's baseline study and monitoring plan which is included in the VCS-PD and other relevant supporting documents against relevant criteria.

The scope of work covered in the validation is described below:

- ✓ To validate whether the project activity meets the requirements of VCS Standard Version 4.5, including additionality, proof of title and compliance with local laws.
- ✓ To evaluate whether the baseline and monitoring plan are in conformance with the applied methodology from the VCS approved GHG program.
- ✓ To confirm that the information presented are completed, consistent, transparent, and free of omission or material error.
- ✓ Background investigation and follow up interviews
- ✓ Issuance of draft validation report with CARs, CLs & FARs, if any
- ✓ Final validation opinion

The information in the VCS PD is reviewed against the criteria of VCS Standard 4.5 and the approved baseline and monitoring CDM methodology AMS-III.D., Version 21.0, and AMS-I.D. Version 18.0. EcoLance has performed validation based on a risk-based approach focusing mainly on the significant risks to meet the qualification criteria and the ability to generate Verified Carbon Units (VCUs). The validation is not meant to provide any consulting towards the client. However, stated request for clarifications and/or corrective actions may provide input for improvement of the project design.

1.3 Reasonableness of Assumptions

All the revisions of the validation report before being submitted to the client were subjected to an independent internal technical review to confirm that all validation activities had been completed according to the pertinent EcoLance's instructions. The technical review was performed by a technical reviewer(s) qualified in accordance with EcoLance's qualification scheme for VCS and CDM validation and verification.

The assumptions, limitations, and methods applied for this project is found to be reasonable, with respect to material errors, omissions and misrepresentations.

1.4 Summary Description of the Project

The project is a biogas-to-energy project that generates renewable energy by capturing biogas from animal manure -via anaerobic digestion- and utilizing biogas systems to produce thermal and electric energy. The project is being implemented by ABRAK ENERJİ ELEKTRİK ÜRETİM ANONİM ŞİRKETİ, as the Project Owner, in Elmadağ district, Ankara province, in Turkey. The central coordinates of the project location are longitude of +40.037864 and latitude of +33.114366.

The location of the project activity mentioned in the VCS PD is checked through Google Map, found consistent and it is accepted. The status of the project activity is verified through a monitoring system and confirmed. The manure which is currently supplied to the proposed project plant is coming from the neighboring farms daily which are fed into the anaerobic digesters at the project site. The start date of the project activity is 31/12/2020, which is date the project has started to reduce GHG emission reductions, is verified through the provisional acceptance certificate obtained by the Ministry of Energy and Natural Sources to generate electricity.

The manure which is currently supplied to the proposed project plant is coming from the neighboring farms are collected daily through special trucks equipped with close tanks to prevent any leakages and are fed into the anaerobic digesters at the project site.

The project has two biogas engines of 1.203 MWe, thus the total installed capacity is 2.406 MWe. The biogas that is released during the biodegradation of organic wastes is used for electricity production in the biogas engine units. Heat released in these units is captured and used at the facility to warm the digesters or used for auxiliary consumption in the administrative buildings. The electricity generated is partly used for auxiliary consumption and the rest is fed to the national grid.

Technical specifications of the equipment involved in the project activity as reported under Section 1.11 of the VCS-PD are verified through the visit during the site visit and are found to be consistent with the same as mentioned in the VCS-PD.

The start date of the project activity is 31.12.2020, which is date of commissioning for the project activity, as verified through the provisional acceptance certificate obtained by the Ministry of Energy and Natural Sources to generate electricity.

The project has obtained the approval of Environmental Impact Assessment (EIA) issued by the Ministry of Environment, Urbanization and Climate Change (MEUC) in Turkey 01.03.2017. PP has submitted the supporting evidence for the EIA issued by MEUC & found consistent.

This project adopts the renewable crediting period of 7 years, twice renewable for a total of 21 years. Its first crediting is from 31.12.2020 to 30.12.2027 (the start and end dates included). The expected average annual emission reductions are 32,857 tCO₂eq/y. Accordingly, the project is expected to generate 229,999 tCO₂eq emissions reduction throughout its first crediting period.

Based on the assessment of the documents, the assessment team can confirm that the project activity is fully functional and implemented as described in the VCS-PD. During the validation process 6 CLs, 2 CAR and 1 FAR are raised and assessed.

2 VALIDATION PROCESS

Through a contract dated 11/10/2022, ABRAK ENERJİ ELEKTRİK ÜRETİM ANONİM ŞİRKETİ has commissioned EcoLance Pvt. Ltd to carry out the validation of the “Elmadag Biogas Power Plant” (hereafter referred to as the Project). The project is undergoing

validation under VCS, the approach adopted to ensure the quality of emission reductions is described in the following sections.

2.1 Method and Criteria

Validation was conducted following the EcoLance's procedures created in line with the requirements of the VCS Standard 4.5, CDM Validation and Verification Standard, and relevant UNFCCC requirements and standard auditing techniques.

A competent assessment team is selected prior to the start of the verification. The team is selected to cover the technical area(s), sectoral scope(s) and relevant host country experience for evaluating the VCS project activity. Additionally, a competent Technical Reviewer or Technical Reviewer Team is appointed to conduct checks on quality and completeness.

The verification team and the technical reviewers consist of the following personnel:

Role/Qualification	Last Name	First Name	Country
VCS Team Leader	Kewat	Shailendra	India
Host country expert	Kaya	Recep	Turkey
Independent Technical Reviewer	Buragohain	Champak	India

The validation approach consists of two phases. In the first phase, EcoLance completed a strategic review; and risk assessment of the project's activities and processes to gain a full understanding of:

- Activities associated with all the sources contributing to the project emissions and emission reductions, including leakage (if relevant);
- Project details and PD;
- Application of Methodology;
- Estimated GHG emission reductions and removals;
- Protocols used to estimate or measure GHG emissions from;
- Collection and handling of data;
- Controls on the collection and handling of data; and
- Means of verifying reported data

At the end of this phase, EcoLance produced a Validation Checklist which, based on the risk assessment of the parameters and data collection and handling processes for each of those parameters, describes the validation approach and the evidence gathering plan.

In the second phase using the Validation checklist, EcoLance verified the implementation of the monitoring plan and the data presented in the VCS PD for the period in question. This involved interview of project proponent representatives, a desk review of the Project description, and site visit. This validation report describes the findings of this assessment.

The key milestones of the validation process have been provided below:

Milestone	Date
Kick-off meeting	26.11.2022
Desk Review	05.12.2022
Site visit	23.12.2022 & 24.12.2022

2.2 Document Review

The validation team has conducted the document review using the VCS Standard 4.5 and the methodologies AMS-III.D., Version 21.0, and AMS-I.D. Version 18.0 as well as applied CDM Tools are the reference criteria.

Prior to the site visit, it was verified by the validation team that the project was listed in the VCS pipeline in line with the VCS requirements Ref:

<https://registry.verra.org/app/projectDetail/VCS/3890>

The VCS-PD version 01 was initially reviewed. Upon further assessment, the validation has raised CARs, CLs and FAR to ensure the compatibility of the PD with the aforementioned criteria. The VCS-PD Version 01 and the VCS-PD Version 03 dated 08/12/2023 were assessed as part of the validation. Relevant documents were reviewed to ensure the consistency and accuracy of the reported data in the PD.

The list of documentation that was reviewed during the validation is given in Appendix-1.

2.3 Interviews

The assessment team has carried out interviews in order to verify the information included in the project documentation and to gain additional information regarding the compliance of the project with the VCS requirements. Before and during on-site visit, the assessment team has interviewed the representatives of the PP to confirm selected information and to clarify issues identified during the document review. The validation team has also conducted interviews with the project authorities to gain comprehensive knowledge about the project activity. The names and designations of the personnel interviewed are mentioned in below.

The main topics covered during the interview are as follows:

- ✓ General Aspects of the project
- ✓ Project Implementation
- ✓ Equipment and operation
- ✓ Staff Training procedures
- ✓ Calibration procedures
- ✓ Monitoring & Measuring System
- ✓ Data collection, recording and archiving procedure
- ✓ QA/QC procedures
- ✓ VCS documentation
- ✓ Emission reduction calculations

Date	Name and Role	Organization	Topic
23/12/2022	Başar Beyazoğlu / Project Owner	ABRAK ENERJİ ELEKTRİK ÜRETİM ANONİM ŞİRKETİ	Description of the project activity, project technology, operation and maintenance, project approval and implementation status, investment Analysis, baseline and additionality, monitoring plan and monitoring arrangements, roles and responsibilities, emergency procedures, VCU ownership.

23/12/2022	Dr. Umut Önder / Consultant	CS CLIMATE SOLUTIONS DANIŞMANLIK ANONİM ŞİRKETİ	Eligibility criteria, baseline and additionality, applicability conditions, emission reductions calculations, additionality and investment analysis, starting date and crediting period. Social & Environmental impacts, VCU ownership.
23/12/2022	Ali Savaş Büken / Site Manager (engineer)	ABRAK ENERJİ ELEKTRİK ÜRETİM ANONİM ŞİRKETİ	Description of the project activity, project technology, operation and maintenance, project approval and implementation status, investment analysis, baseline and additionality, monitoring plan and monitoring arrangements, roles and responsibilities, emergency procedures.
23/12/2022	Samet Ay / Laboratory responsible	ABRAK ENERJİ ELEKTRİK ÜRETİM ANONİM ŞİRKETİ	Project implementation & operational status, monitoring of the lab operations, technologies, measurement, and calibration procedures.
23/12/2022	İsmail Bostan / Electrical & Mechanical Technician	ABRAK ENERJİ ELEKTRİK ÜRETİM ANONİM ŞİRKETİ	Monitoring of the operational parameters, maintenance and calibration procedures, knowledge about the equipment and instruments in the plant.
23/12/2022	Baran Saltürk / SCADA operator	ABRAK ENERJİ ELEKTRİK ÜRETİM ANONİM ŞİRKETİ	Monitoring of the operational parameters, calibration procedures, knowledge about the equipment and instruments in the plant, operation of the SCADA, and data export procedures.
23/12/2022	Yılmazlar Besi Çiftliği	Farm Owner / Kerim Yılmaz	Manure treatment system followed at the farm, type & number of animals in the farm, amount of manure generated daily, manure supply agreement, baseline survey conducted, records maintained at the farm, VCU ownership.
23/12/2022	Ali Bey Çiftliği	Farm Owner / Necmettin Erden	

2.4 Site Visits

The validation site visits were conducted on 23/12/2022 and 24/12/2022. Site visit included the inspection of central plant and the visit of the livestock farms supplying manure to the plant. A ground visit of the project plan was conducted during the site visit and the validation and verification team interviewed project implementer representative, operation staff and farm owners. The list of interviewed people has been given in Section 2.3. During the visits, the accuracy of project description and the baseline scenario was evaluated, the technology implemented for the project activity was checked, and the supporting documents related to the project activity were examined.

No	Farm Name	LATITUDE	LONGITUDE	Distance (km)
1	Ali Bey Çiftliği (Necmettin Erden)	40.03514	33.073257	7.3
2	Erol Yılmaz	39.96814	33.2067	10.6
3	Bey Besi (Veysel Karabey)	40.0221	33.05187	15.3
4	Şenol Yılmaz Besi Çiftliği	39.95783	33.2177	12.2
5	Saygın Bey (Veysel Karabey)	40.04001	33.11034	3.6
6	Aytekin Besi	39.96402	33.22377	17.1
7	Has Besi	39.95732	33.21872	12.3
8	İzzet Alp	39.97836	33.01407	26.8
9	Yılmazlar Besi (Kerim Yılmaz)	39.90155	32.89994	23.1

The VVB has checked the manure supply agreements and the baseline surveys. It was also confirmed during the site visits that the geo-coordinates of the manure supplying farms provided in the VCS-PD are accurate. The periodicity of the manure collection activity from the contracted farms were also confirmed upon weighbridge recordings and cross-checked via interviews conducted with farms.

Please, see the details of the interviews with the project owner, other relevant stakeholders such as the site manager, laboratory staff, technical staff, and farm owners in Sections 2.2 and 2.3 above. These interviews were conducted to confirm the provided details of the project design, information given about the project, and results reported in the VCS PD.

2.5 Resolution of Findings

The objective of this phase of the validation is to resolve any outstanding issues, which need to be clarified for EcoLance's positive conclusion on the project design.

To guarantee transparency a validation protocol has been customized for the project. The protocol shows in a transparent manner the requirements, means of validation and the results from validating the identified criteria. The findings raised during this phase has been summarized in the table below. The completed validation protocol is enclosed in Appendix 2 of this report.

A corrective action request (CAR) is raised if one of the following occurs:

- The project participants have made mistakes that will influence the ability of the project activity to achieve real, measurable additional emission reductions.
- The VCS Version 4.5 requirements have not been met.
- There is a risk that the emission reductions cannot be monitored or calculated.

A clarification request (CL) is raised if information is insufficient or not clear enough to determine whether the applicable VCS Version 4.5 requirements have been met.

The final VCS PD Version 3 submitted by project proponents on 08/12/2023 serves as the basis for the final assessment presented. Additional changes to the project during the validation process are not considered to be significant with respect to the main VCS objectives. The two CDM/VCS main objectives are the reduction of anthropogenic GHG emissions and the contribution of sustainable development to the host country.

Areas of validation findings	No. of CL	No. of CAR	No. of FAR
Project design document	00	00	00
Description of project activity	02	01	00
Application of selected baseline and monitoring methodology and selected standardized baseline			
- Applicability of methodology and standardized baseline	00	00	00
- Deviation from methodology	00	00	00
- Deviation from project description	00	00	01
- Clarification on applicability of methodology, tool and/or standardized baseline	00	00	00

- Demonstration of additionality	00	00	00
- Emission reductions	01	00	00
- Monitoring plan	01	00	00
-Stakeholders consultation process	00	00	00
- Public comments	00	00	00
Others (please specify)-project boundary	01	00	00
Others (please specify)-Matter related to Emission reduction calculation- ER achieved – Actual ER achieved calculations,	00	00	00
Other (Project details)-Parameters available at validation and Monitored Parameters.	01	01	00
Total	06	02	01

The list of findings and their resolution is presented in appendix 2 of this report.

2.5.1 Forward Action Requests

One Forward Action Request (FAR) is raised during validation activities regarding the assessment and reporting of any manure supplier farms that might be included in the project activity at the upcoming verification periods.

3 VALIDATION FINDINGS

3.1 Project Details

As per the project type defined by the UNFCCC and VCS standard version 4.5 Guidelines, the project activity falls under sectoral scope 01: Energy industries (renewable -/ non-renewable sources) (Type I Component), and sectoral scope 13: Waste handling and disposal (Type III Component).

Elmadag Biogas Power Plant Project (hereafter referred to as the project) is a biogas-to-energy project that will generate renewable energy by capturing biogas from animal manure -via anaerobic digestion- and utilizing it to produce thermal and electric energy through gas engines. The project is being implemented by ABRAK ENERJİ ELEKTRİK ÜRETİM ANONİM ŞİRKETİ, the Project Owner, in Elmadağ district, Ankara province, in Turkey. The central coordinated of the project location are longitude of +40.037864 and latitude of +33.114366. The location of the project activity mentioned in the VCS-PD is checked via the Google Maps and KMZ file, found consistent and is accepted. The project has a total installed capacity of 2.406 MWe which was confirmed via checking the production license was issued by Energy Market Regulatory Authority (EMRA) on 01/08/2019. EIA decision was taken for the project

activity in 01/03/2017 with the decision number 4540. The project has a total installed capacity of 2.406 MWe. This project adopts the renewable crediting period of 7 years, twice renewable for a total of 21 years.

Upon site visits and document review, the type of four anaerobic digesters installed at the facility is confirmed as EPDM-membrane enclosed anaerobic digester. For the electricity generation, two Jenbacher gas engines with a total capacity of 2.406 MWe is installed at the facility which was confirmed through site visits and document review. Flare, which is installed as an emergency equipment, is confirmed to have a 1100 kW thermal capacity and a height of 5.7 m.

The PP has entered into a long-term power purchase agreement (PPA) with the grid authority. The copy of PPA is validated to confirm the same. The output capacity of the project activity is validated by physical inspection, validation of PPA and commissioning certificate.

Upon checking the baseline surveys filled by farm owners, along with the observations conducted at the farms during the interviews, it was verified that prior to project activity, the manure generated at neighbouring animal farms were left to decay in anaerobic conditions within uncovered anaerobic lagoons. This period in average takes about 2-3 months, then the decayed manure directly applied to land without any proper treatment. This led to generation and release of GHGs directly into the atmosphere.

The purpose of the project activity is to treat the animal manure generated from farms of Elmadag district, Ankara province, in Turkey. It aims to capture methane from animal manure through anaerobic fermentation and generate electricity through biogas recovery, the electricity generated at the plant is partly fed to the national grid, avoiding and/replacing the dispatch of the same amount of electricity from fossil-fuel-based power plants connected to the national grid. This acts as a solution to reduce pollution and achieve reduction in greenhouse gas (GHG) emissions.

The estimated net annual emission reductions are 32,857 tCO₂eq/y. Accordingly, the project is expected to generate 229,999 tCO₂eq emissions reduction throughout its first crediting period.

The technical specification of the project activity equipment's has been checked through the inspections of all the equipment's installed at site and are found to be consistent with the mentioned under section 1.11 of VCS-PD.

This is not a grouped project activity. Thus, this section is not applicable for this project. By checking project approval and site visit, assessment team confirmed the project has been designed to include a single installation of an activity.

Project proponent

The assessment team confirms that the parties below act as project proponents through checking the contractual agreement between Climate Solutions Danışmanlık A.Ş and Abrak Enerji Elektrik Üretim Anonim Şirketi. The project proponents and their titles are clearly defined under Section 1.5 and 1.6 of the VCS PD. Project is being developed by CS CLIMATE SOLUTIONS DANIŞMANLIK ANONİM ŞİRKETİ and ABRAK ENERJİ ELEKTRİK ÜRETİM ANONİM ŞİRKETİ. CS CLIMATE SOLUTIONS DANIŞMANLIK ANONİM ŞİRKETİ is the project consultant and ABRAK ENERJİ ELEKTRİK ÜRETİM ANONİM ŞİRKETİ is the project owner. Same has been confirmed by the commissioning certificate and the PPA, which is on the name of ABRAK ENERJİ ELEKTRİK ÜRETİM ANONİM ŞİRKETİ.

Project start date

As per VCS Standard V4.5, clause 3.8.1: non-AFOLU projects shall complete validation within two years of the project start date. Wherein, the project start date of a non-AFOLU project is the date on which the project began generating GHG emission reductions or removals.

The starting date of project activity is 31.12.2020 as confirmed through the date of commercial operation of the proposed project activity, which it is in line with the VCS Standard.

As per the VCS Standard, the project start date is the date on which the project began generating GHG emission reductions or removals.

Project crediting period

The crediting period of the project activity is renewable 7 years, which can be renewed twice. The first crediting period is 7 years from 31.12.2020 to 30.12.2027 (both days included).

Project scale and estimated GHG emission reductions or removals

The estimated average annual emission reductions from the project are estimated average net annual emission reduction is 32,857 tCO₂ per year over the 7 years crediting period.

Projects are categorized by size according to their estimated average annual emission reductions or removals according to VCS Standard v4.5. Since the estimated annual

emission reductions resulting from the project is less than 300,000 tCO₂, the category of the project is defined as “Project” as per the VCS Standard.

Estimated GHG emission reductions of the project activity over the first crediting period is given below:

Project Scale	
Project	√
Large project	

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
31/12/2020	90
01/01/2021-31/12/2021	32,857
01/01/2022-31/12/2022	32,857
01/01/2023-31/12/2023	32,857
01/01/2024-31/12/2024	32,857
01/01/2025-31/12/2025	32,857
01/01/2026-31/12/2026	32,857
01/01/2027-30/12/2027	32,767
Total estimated ERs	229,999
Total number of crediting years	7
Average annual ERs	32,857

Project location

The Project location is discussed in section 1.12 of the VCS-PD. The project location is taken as per the site layout as discussed below.

The project is located in Elmadag district, Ankara province, in Turkey. The central coordinated of the project location are longitude of +40.037864 and latitude of

+33.114366. Project compliance with applicable laws, statutes and other regulatory frameworks.

The two of main regulatory approvals for such project types to prove that they are in compliance with relevant laws, statues and regulatory frameworks in Turkey are the Environmental Impact Assessment and the Electricity Generation License. The project has obtained the approval of Environmental Impact Assessment (EIA) issued by the Ministry of Environment, Urbanization and Climate Change in Turkey on 01.03.2017 and the Electricity Generation License (EGL) issued by the Energy Market Regulatory Authority (EMRA) in Turkey on 01.08.2019. The project is in compliance with the given laws.

Ownership

Right of use: The legal right to control and operate the project activity belongs to ABRAK ENERJİ ELEKTRİK ÜRETİM ANONİM ŞİRKETİ. This was also confirmed during the interview and upon checking the agreement between the ABRAK ENERJİ ELEKTRİK ÜRETİM ANONİM ŞİRKETİ and Electricity supply licence, which ensures that the emission reduction credits from the project go to the project owner. Apart from this VVB has checked Business license of the project owner and the purchasing contract of generators and the purchasing power agreement to confirm the ownership.

Emissions trading programs and other binding limits:

The project does not reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading.

Participation under other GHG programs:

The project has not been registered or seeking registration under other GHG programs. Same has been confirmed by verifying different registries, The details of the registries checked are as follows:

- <https://www.climateactionreserve.org/>
- <https://registry.goldstandard.org/projects?q=&page=1>
- <https://iceland.itmoregistry.net/Public/Project>
- <https://biocarbonregistry.com/en/projects/>
- https://projects.globalcarboncouncil.com/pages/submitted_projects

Other forms of credit and supply chain (Scope 3) emissions:

It was confirmed that the project has not created another form of Environmental Credit or renewable energy certificates. Moreover, the project has not sought or received another form of GHG-related credit, including renewable energy certificates.

When Supply Chain (Scope 3) emissions are assessed in line with the VCS Standard 4.5 and Corporate Value Chain (Scope 3) Accounting and Reporting Standard¹, the assessment team confirms that the project activity is not related with Scope 3 emissions since it does not directly impact emissions associated with any good or service other than the electricity supplied to the national grid. However, necessary precautions were taken against double counting as confirmed by the validation team.

Sustainable Development Conditions:

The project activity has been contributed to five of the UN Sustainable Development Goals (SDGs). These SDGs are given below:

-SDG 7 Affordable & Clean Energy: Ensure access to affordable, reliable, sustainable, and modern energy for all, PP indicates contribution to the Target 7.2, indicator 7.2.1 which is defined as “Renewable energy share in the total final energy consumption”. The validation team confirms that the project activity contributes to this goal by generating 2.406 Mwe to feed it to the national grid after using it for auxiliary consumption. The project developer’s operational logbooks and total capacity of the biogas-to-energy plant has been checked to verify this contribution. The contribution to this goal will also be monitored and reported during the monitoring periods. (SDG 7.2.1).

-SDG 8 Decent Work & Economic Growth: Promote sustained, inclusive, and sustainable economic growth, full and productive employment, and decent work for all. The project is expected to create some permanent jobs for maintenance and operational activity of the project (SDG 8.0). The validation team also confirms that the reported self-defined SDG contribution is in line with the requirements of VERRA Sustainable Development Contributions Report Template. This parameter will be monitored via the employment records of the facility.

¹ https://ghgprotocol.org/sites/default/files/standards/Corporate-Value-Chain-Accounting-Reporting-Standard_041613_2.pdf

-SDG 12 Responsible Consumption & Production: “Integrate climate change measures into national policies, strategies, and planning”, PP indicates contribution to a self-defined SDG indicator 12.0 defined as “amount of amount of waste recycled/recovered by the project activity”. It is reported that animal waste collected from the neighbouring farms are recycled/recovered to make a meaningful contribution to the economy and environment. The validation team confirms that this self-defined indicator is indeed created in line with the VCS SDG contribution guidelines and makes a valuable contribution to the United Nations’ 2015 call to action. The contribution to this goal will also be monitored and reported during the future monitoring periods through the examination of weighbridge logbooks (SDG 12.0).

-SDG 13 Climate Action:

“Take urgent action to combat climate change and its impacts”, PP indicates contribution to a self-defined SDG indicator 13.0 defined as “total GHG emission reductions per year”. PP reports that the project activity is expected to generate 32,857 metric tons of CO₂eq/y. The validation team confirms that this self-defined indicator is indeed created in line with the VCS SDG contribution guidelines and makes a valuable contribution to the United Nations’ 2015 call to action. This contribution will be monitored and reported during the future monitoring periods through GHG ER calculations.

-SDG 17: “Strengthen the means of implementation and revitalize the Global Partnership for Sustainable Development”, PP indicates contribution to the SDG Target 17.17, Indicator 17.17.1 defined as “Amount in United States dollars committed to public-private partnerships for infrastructure”. PP reports that the project activity is in a contractual agreement with the Turkish state under the Public-Private Partnership-based (PPP) Renewable Energy Support Scheme (YEKDEM) aiming at increase in electricity generation from renewable sources in Turkey which has enabled the project to receive USD 0.133 per KWe fed to the national grid. The validation team confirms this agreement upon checking the supporting document and interviewing the project manager (SDG 17.17.1).

Rejection by other GHG programs: The project has not been rejected by other GHG programs.

Additional information relevant to the project

Eligibility criteria for grouped projects:

The project is not a grouped project.

Commercially sensitive information:

No commercially sensitive information has been excluded from the public version of the project description.

Any further information:

Not applicable.

Overall, it is confirmed that the PD is accurate, complete, and provides an understanding of the nature of the project.

3.2 Safeguards

3.2.1 No Net Harm

The Environmental Impact Assessment (EIA) Report has been approved and issued by the Ministry of Environment, Urbanization and Climate Change in Turkey on March 1st, 2017. In accordance with the Turkish laws and regulations, EIA's approval shall be only made if a project subjected to the approval does not make any negative environmental and socio-economic impacts. Considering the fact that the proposed project activity already obtained EIA approval by the Ministry of Environment, Urbanization and Climate Change in Turkey, it is therefore possible to claim that there is no net harm linked to the project.

All potential environmental and socio-economic impacts have been discussed in the EIA, the project does not pose any harm to its environment. The EIA report clearly defines possible negative effects and the control measures taken to prevent such outcomes. As described in the VCS-PD, possible impacts are found to be appropriately managed thus causing no harm. The assessment team checked the same and find that there are no negative potential environmental and socio-economic impacts detected during the site visit. Moreover, based on the professional experience of assessment team, and site visit, the assessment team confirm that there are no negative potential environmental and socio-economic impacts, thus there is "no net harm".

3.2.2 Local Stakeholder Consultation

As per the VCS requirements, it is necessary to invite the relevant stakeholders, prior of the validation process. The assessment team checked the relevance of the dates during the validation interview with the farm owner and project representatives.

Local stakeholder consultation was conducted through distributing questionnaires to local stakeholders on 06.07.2022 by the project owner. Considering the cases due to the Covid-19 pandemic in the world, it was decided to hold the meeting online.

Furthermore, through interviews, the assessment team confirmed that communications with stakeholders were conducted as per Section 3.17 of VCS Standard 4.3². Key implementation schedules or changes of the project have been and will be communicated to the local authority, who will inform the neighborhood committee and the local residents. The comments and suggestions from residents will be collected by the local authority and feedback to the project owner will be sent.

By checking the questionnaires, the assessment team confirms that the local stakeholder has no negative comments for the construction of the project activity.

Furthermore, a grievance book is put in the communication room at the project owner company. The local stakeholders can provide feedback through this way. Also, the village head will reach PO if any comments are received. The assessment team confirms that input/grievance mechanism put into place through the connection between local stakeholders, mukhtars, project proponent, and project owner provides a suitable way for on-going communication.

Thus, the Validation team is of the opinion that the stakeholder meeting was adequate and appropriate. There were no negative comments raised by the stakeholders and they were totally in support of the setting up of these kinds of projects in the region.

3.2.3 Environmental Impact

The Environmental Impact Assessment (EIA) certificate has been approved and issued by the Ministry of Environment, Urbanization and Climate Change in Turkey on March 1st, 2019. In accordance with the Turkish laws and regulations, EIA's approval shall be only made if a

² The validation team was checked and confirmed that the most recent standard was considered in time of Local Stakeholder Consultation.

project subjected to the approval does not make any negative environmental and socio-economic impacts. Considering the fact that the proposed project activity already obtained EIA approval by the Ministry of Environment, Urbanization and Climate Change in Turkey, it is therefore possible to claim that there is no net harm linked to the project. The validation team checked the EIA certificate and accepted.

3.2.4 Public Comments

This project was open for public comment from November 10th, 2022, to December 10th, 2022, on its official VERRA/VCS webpage³.

The detail was checked by the assessment team in the following web platform:
<https://registry.terra.org/app/projectDetail/VCS/3890>

During the period, no public comments were received.

3.2.5 AFOLU-Specific Safeguards

Since the project is a non-AFOLU project, this section is not applicable.

3.3 Application of Methodology

3.3.1 Title and Reference

The assessment team checked that following methodologies and tools are applicable for the project activity. The details are as below:

Title: AMS-III.D. Methane recovery in animal manure management systems, v21.0

Reference: The project activity meets the eligibility criteria of large-scale project

Sectoral Scope: 13.

Category: Approved Consolidated Methodology

Title: AMS-I.D. Grid connected renewable electricity generation, v18.0

Reference: The project activity meets the eligibility criteria of large-scale project

Sectoral Scope: 01.

Category: Approved Consolidated Methodology

Tools referred with above methodology and applicable for project activity are:

- “Tool to calculate project or leakage CO₂ emissions from fossil fuel combustion”, Version 03.0;
- “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation”, Version 03.0;

³ <https://registry.terra.org/app/projectDetail/VCS/3890>

- “Project emissions from flaring”, Version 04.0;
- “Project and leakage emissions from anaerobic digesters”, Version 02.0;
- “Tool to calculate the emission factor for an electricity system”, Version 07.0;

The VVB has checked and confirmed that the title and version of the mentioned tools in the VCS-PD are correct and valid.

3.3.2 Applicability

The AMS-III.D “Approved baseline and monitoring methodology – Methane recovery in animal manure management system.” Version 21.0⁴ applies to project activities involving the replacement or modification of anaerobic animal manure management systems in livestock farms to achieve methane recovery and destruction by flaring/combustion or gainful use of the recovered methane. It also covers treatment of manure collected from several farms in a centralized plant.

The methodology is applicable under the following conditions:

Applicability of AMS-III.D. – v21.0					
Ref.		Application Criteria		Applicability	
The methodology is applicable under the following conditions:					
1.		(a)		The livestock population in the farm is managed under confined conditions;	
				The condition of manure supplier farms was checked by the VVB during the site visits and interviews. The VVB confirms that livestock population is managed under confined conditions in line with the relevant Turkish laws and regulations confining conditions, especially regarding animal health, such as Law No. 5996 ⁵ , Law No. 3285 ⁶ and Regulation on Livestock Farms ⁷ . These farms are constantly inspected by the relevant governmental	

⁴ <https://cdm.unfccc.int/methodologies/DB/H9DVS2407GEZQYLYNWUX23YS6G4RCv>

⁵ https://www.tarimorman.gov.tr/Belgeler/ENG/Legislation/law_veterinary_services.pdf

⁶ http://www.zmo.org.tr/mevzuat/mevzuat_detay.php?kod=55

⁷ <http://www.resmigazete.gov.tr/eskiler/2006/08/20060809-4.htm>

			institutions to check whether they are managed under confined conditions set by the relevant Turkish laws and regulations. Hence this condition is applicable.
	(b)	Manure or the streams obtained after treatment are not discharged into natural water resources (e.g. river or estuaries), otherwise "AMS-III.H Methane recovery in wastewater treatment" shall be applied;	Manure is left for decaying under anaerobic conditions and is not discharged into natural water resources. VVB has verified this via site visits, baseline survey data, and interview with the farm owners. Further, the streams obtained after treatment are used as fermented product rather than discharged into natural water resources. The VVB has confirmed this through site visit to the central plant. Hence this condition is applicable.
	(c)	The annual average temperature of baseline site where anaerobic manure treatment facility is located is higher than 5°C;	The annual average temperature of baseline site where manure is collected from multiple farms in ANKARA is 12.0°C ⁸ as confirmed by data statistics. Hence, the proposed project activity complies with this criteria/condition.

⁸ <https://mgm.gov.tr/veridegerlendirme/il-ve-ilceler-istatistik.aspx?m=ANKARA> accessed on 20.10.2022

	(d)	In the baseline scenario the retention time of manure waste in the anaerobic treatment system is greater than one month, and if anaerobic lagoons are used in the baseline, their depths are at least 1 m;	The depth of open anaerobic lagoons in the farms included in the project activity is greater than 1 m, as given in the Appendix-1 of the PD. VVB checked it via photographs, baseline survey data, video and verified during the site visits. Hence applicable.
	(e)	No methane recovery and destruction by flaring or combustion for gainful use takes place in the baseline scenario	Through the interviews with the farm owners and site visits, it is confirmed that animal manure waste was left to decay in open anaerobic lagoon at the livestock farm and the emission of methane that would be emitted to the atmosphere directly without any methane recovery and destruction facility in the baseline scenario and no methane recovery and destruction by flaring or combustion for gainful use takes place in the baseline scenario. Therefore, condition is applicable.
2.	(a)	The residual waste from the animal manure management system shall be handled aerobically, otherwise the related emissions shall be taken into account as per relevant procedures of "AMS-III.AO Methane recovery through controlled anaerobic digestion". In the case of soil application, proper conditions and procedures (not resulting in methane emissions) must be ensured;	By checking EIA Report of the project activity, and interviewing with PP during site visit, and through site visit, it is confirmed that the residual waste from the animal manure management system of the project activity is sent out and used for agriculture fertilization aerobically in the nearby farmlands. Therefore, condition is applicable.
	(b)	Technical measures shall be used (including a flare for exigencies) to ensure that all biogas produced by the digester is used or flared;	By checking EIA Report of the project activity, as well as site visit, it is confirmed that for electricity generation, two Jenbacher gas engines with a total capacity of 2.406 MWe is installed for utilizing the generated biogas. Further, an emergency flare was installed at the

			project site to ensure that biogas is combusted when exigencies happened. Therefore, condition is applicable.
	(c)	The storage time of the manure after removal from the animal barns, including transportation, should not exceed 45 days before being fed into the anaerobic digester. If the project proponent can demonstrate that the dry matter content of the manure when removed from the animal barns is larger than 20%, this time constraint will not apply	By interviewing with PP during site visit, and through site visit, it is confirmed that the storage time of the manure after removal from the animal barns, including transportation, are less than 24 hours, does not exceed 45 days before being fed into the anaerobic digester. Therefore, condition is applicable.
3.		Projects that recover methane from landfills shall use "AMS-III.G Landfill methane recovery" and projects for wastewater treatment shall use AMS-III.H. Projects for composting of animal manure shall use "AMS-III.F Avoidance of methane emissions through composting". Project activities involving co-digestion of animal manure and other organic matters shall use the methodology "AMS-III.AO Methane recovery through controlled anaerobic digestion";	By checking EIA Report and through site visits it is confirmed that the project recovers methane from anaerobic animal manure management system installed at the central plant, not from landfills, or wastewater treatment, or composting of animal manure, or co-digestion of animal manure and other organic matters. Therefore, this criterion is applicable.
4.		Utilization of the recovered biogas in one of the options detailed in AMS-III.H is also eligible under this methodology. The respective procedures in AMS-III.H shall be followed in this regard. If the recovered biogas is used to power auxiliary equipment of the project activity, it should be taken into account accordingly, using zero as its emission factor; however, energy used for such purposes is not eligible as an SSC CDM Type I project component	Through site visits and checking the EIA report, the VVB has confirmed that the project does not utilize biogas from wastewater treatment systems. Hence, only AMS-III.D is applicable.
5.		New facilities (Greenfield projects) and project activities involving capacity	The VVB has checked the suitability of project activity against the relevant

	additions compared to the baseline scenario are only eligible if they comply with the related and relevant requirements in the "General guidelines for SSC CDM methodologies"	requirements of the "General guidelines for SSC CDM methodologies Version 23⁹". Such requirements state that "The sum of the size of components of a project activity belonging to the same type (capacity for Type I, energy savings for Type II and emission reductions for Type III) should not exceed the limits for small-scale CDM project activities (e.g. the limit for methane recovery component is 60 ktCO₂e/yr and the limit for the electricity production component is 15 MW output capacity)" which is confirmed as suitable since the project capacity is 1.5MWe/h which and the estimated annual emission reductions that is generated from the implementation of the proposed project activity is 32,857 tCO₂e/y, which is lower than the threshold of 60,000 tCO₂e/y set in the Guideline. Therefore, the proposed project activity complies with the Guideline. In addition, the VVB has confirmed that the step-wise procedure to identify the baseline scenarios of the Guideline, "Procedure to determine baseline scenario for Type II and Type II greenfield projects" is followed properly. Section 3.4. of the PD demonstrates the relevant steps where all the alternatives of manure disposal presently followed by the farms are presented and the one applicable baseline scenario has
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⁹ https://cdm.unfccc.int/sunsetcms/storage/contents/stored-file-20190916153417994/MethSSC_guid25.pdf

		been identified. Hence, the project activity is suitable for this criterion.
6.	The requirements concerning demonstration of the remaining lifetime of the replaced equipment shall be met as described in the "General guidelines for SSC CDM methodologies"	By checking the project feasibility report and supporting documents, the VVB confirms that the project is a Greenfield project, does not involve replacement of equipment. Hence, this condition is not applicable to the proposed project activity.
7.	Measures are limited to those that result in aggregate emission reductions of less than or equal to 60 kt CO ₂ equivalent annually from all Type III components of the project activity	The generation of the project's emission reductions is estimated as 32,857 tCO ₂ e/y, which is lower than the threshold of 60,000 tCO ₂ e/y. Therefore, the proposed project activity complies with this condition.

Applicability of AMS-I.D. – v18.0			
Ref.	Application Criteria		Applicability
The methodology is applicable under the following conditions:			
1.	(a)	Install a Greenfield plant;	The VVB has checked the project installation documents and the feasibility report to confirm that the project activity is implemented as a greenfield project.
	(b)	Involve a capacity addition in (an) existing plant(s);	Since the project activity is a greenfield plant, this criterion is not applicable.
	(c)	Involve a retrofit of (an) existing plant(s);	Since the project activity is a greenfield plant, this criterion is not applicable.
	(d)	Involve a rehabilitation of (an) existing plant(s)/unit(s); or	Since the project activity is a greenfield plant, this criterion is not applicable.

	(e)	Involve a replacement of (an) existing plant(s)	Since the project activity is a greenfield plant, this criterion is not applicable.
2.	(a)	The project activity is implemented in an existing reservoir with no change in the volume of reservoir;	Since the proposed project activity is not a hydro power plant, this condition is not applicable to the project.
	(b)	The project activity is implemented in an existing reservoir, where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the project emissions section, is greater than 4 W/m ² ;	Since the proposed project activity is not a hydro power plant, this condition is not applicable to the project.
	(c)	The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the project emissions section, is greater than 4 W/m ²	Since the proposed project activity is not a hydro power plant, this condition is not applicable to the project.
3.		If the new unit has both renewable and non-renewable components (e.g. a wind/diesel unit), the eligibility limit of 15 MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel, the capacity of the entire unit shall not exceed the limit of 15 MW	The VVB has confirmed that the project activity only uses a renewable component (biogas) through site visits and checking the feasibility and EIA report.
4.		Combined heat and power (co-generation) systems are not eligible under this category.	Co-generation systems require the installation of a heat engine simultaneously generates both heat and power as per the sister methodology ACM0018, Version 4.0. The VVB has confirmed that the project activity only includes a gas engine for electricity generation by checking the feasibility report and through the site visits.

5.	In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be physically distinct from the existing units	Since the project activity is a greenfield plant, this criterion is not applicable.
6.	In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project, the total output of the retrofitted, rehabilitated or replacement power plant/unit shall not exceed the limit of 15 MW	Since the project activity is a greenfield plant, this criterion is not applicable.
7.	In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid then the baseline for the electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as “AMS-I.C.: Thermal energy production with or without electricity” shall be explored	The project utilizes methane recovered from animal manure, which is eligible under Type III category, to generate electricity and supply it to the grid. AMS-I.D defines the baseline conditions for greenfield projects as “the electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid”. The VVB confirms that the project activity generates electricity from the recovered methane and does not involve heat generation via site visits and checking the project feasibility report. Hence, the criterion is applicable.
8.	In case biomass is sourced from dedicated plantations, the applicability criteria in the tool “Project emissions from cultivation of biomass” shall apply	The VVB has confirmed that the project does not source biogas from dedicated plantations rather generates biogas in anaerobic digesters through the site visit. Thus,

		this condition is not applicable to the project.
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Applicability conditions of "Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation", Version 03.0¹⁰

If emissions are calculated for electricity consumption, the tool is only applicable if one out of the following three scenarios applies to the sources of electricity consumption:

- (a) Scenario A: Electricity consumption from the grid. The electricity is purchased from the grid only, and either no captive power plant(s) is/are installed at the site of electricity consumption or, if any captive power plant exists on site, it is either not operating or it is not physically able to provide electricity to the electricity consumer;
- (b) Scenario B: Electricity consumption from (an) off-grid fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants are installed at the site of the electricity consumer and supply the consumer with electricity. The captive power plant(s) is/are not connected to the electricity grid; or
- (c) Scenario C: Electricity consumption from the grid and (a) fossil fuel fired captive power plant(s). One or more fossil fuel fired captive power plants operate at the site of the electricity consumer.

Emissions are calculated for electricity consumption, Scenario A "Electricity consumption from the grid" applies to the sources of electricity consumption for the project activity.

This tool can be referred to in methodologies to provide procedures to monitor amount of electricity generated in the project scenario, only if one out of the following three project scenarios applies to the recipient of the electricity generated:

- (a) Scenario I: Electricity is supplied to the grid;
- (b) Scenario II: Electricity is supplied to consumers/electricity consuming facilities; or
- (c) Scenario III: Electricity is supplied to the grid and consumers/electricity consuming facilities.

This tool can be referred to in methodologies to provide procedures to monitor amount of electricity generated in the project scenario, Scenario I: "Electricity is supplied to the grid" applies to the recipient of the electricity generated

¹⁰ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-05-v3.0.pdf>

This tool is not applicable in cases where captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage. The tool only accounts for CO₂ emissions.

No captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage. The applicability of this tool is checked and confirmed during the site visits and via the review of project feasibility report.

Applicability conditions of "Tool to calculate the emission factor for an electricity system" Version 07.0¹¹

This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).

The electricity supplied by the project was exported to Turkey national grid. OM, BM and CM are estimated using the tool for calculating baseline emissions for the project activity.

Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in "Appendix 1: Procedures related to off-grid power generation" should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.

The electricity generated by the project activity is exported to Turkey National grid. Hence, the tool is applicable. The emission factor for the project electricity system is calculated for grid power plants.

Although Turkey is an Annex-I country, Turkey is considered in a different situation since Turkey's contribution to GHG emissions is considerably below the average of Annex-I countries.¹² It was also stated by the Ministry of Foreign Affairs of Turkey that "...Türkiye had no historical responsibility for greenhouse gas emissions in 1992 despite being included in

¹¹ <https://cdm.unfccc.int/methodologies/PAMethodologies/tools/am-tool-07-v7.0.pdf>

¹² https://unfccc.int/sites/default/files/resource/cp2021_inf02_0.pdf

Annex-I. Indeed, by that time, Türkiye was in the last row of the Annex I list among 36 countries in terms of greenhouse gas emission with per capita ratio of 3,88 tCO₂e (less than 1% of the global total), while the average greenhouse gas emission of the Annex I countries was 14,37 tCO₂e per capita.”¹³

Based on Turkey's special situation, this tool is considered applicable to the project.

Applicability conditions of “Project emissions from flaring” Version 04.0¹⁴

This tool provides procedures to calculate project emissions from flaring of a residual gas. The tool is applicable to enclosed or open flares and project participants should document in the PD the type of flare used in the project activity.

This tool is applicable to the flaring of flammable greenhouse gases where:

- (a) Methane is the component with the highest concentration in the flammable residual gas; and
- (b) The source of the residual gas is coal mine methane or a gas from a biogenic source (e.g. biogas, landfill gas or wastewater treatment gas).
- (c) The tool is not applicable to the use of auxiliary fuels and therefore the residual gas must have sufficient flammable gas present to sustain combustion. For the case of an enclosed flare, there shall be operating specifications provided by the manufacturer of the flare.

During the interview, the assessment team confirmed that flare is installed in the project site for emergency purposes, and the applicability of the criteria were confirmed based on document submitted by the PP based on the process data, such as the records on the composition of biogas measured by the gas analysers. The VVB also confirms that auxiliary fuels are not used for flaring through the site visits. Therefore, this tool is applicable for the project activity.

Applicability conditions of “Tool to calculate project or leakage CO₂ emissions from fossil fuel combustion” Version 03.0¹⁵

This tool provides procedures to calculate project and/or leakage CO₂ emissions from the combustion of fossil fuels. It can be used in cases where CO₂ emissions from fossil fuel combustion are calculated based on the quantity of fuel combusted and its properties.

¹³ <https://www.mfa.gov.tr/united-nations-framework-convention-on-climate-change- unfccc -and-the-kyoto-protocol.en.mfa>

¹⁴ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-06-v4.0.pdf>

¹⁵ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-03-v3.pdf>

By performing interview and checking the project feasibility report, the VVB confirms that the project consumes diesel oil combusted by the engines of vehicles transporting manure. The bills received from the dedicated vehicles were also taken into account.

Applicability conditions of “Methodological Tool: Project and Leakage Emissions from Anaerobic Digesters”, Version 02.0¹⁶

Assessment team confirms that since the project activity applies the methodological tool “Project and Leakage Emissions from Anaerobic Digesters”, this methodological tool is applicable to project activity.

- CO₂ emissions from consumption of electricity associated with the operation of the anaerobic digester;

Since the PP has chosen the procedure using monitored data (Option 1) for the determination of project emission from the electricity consumption, the assessment team confirms that the relevant condition is applicable to the project.

- CO₂ emissions from consumption of fossil fuels associated with the operation of the anaerobic digester;

During the site visit, the assessment team also confirmed that the fossil fuel is not used during the operation of the anaerobic digesters. Therefore, the relevant condition is not applicable to the project.

- CH₄ emissions from the digester (emissions during maintenance of the digester, physical leaks through the roof and side walls, and release through safety valves due to excess pressure in the digester);

Since the PP has chosen the relevant procedure (Option b) including the applied tool, the assessment team has confirmed that the relevant condition applies to the project.

- CH₄ emissions from flaring of biogas;

During the site visit, the assessment team confirmed that the open flare is installed at the biogas plant and is only used in case of emergency or malfunction in the plant. Besides, if such measurement is positive, the emissions from flaring will be determined following the methodological tool “Project emissions from flaring”, version 03.0. Therefore, the relevant condition does not apply to the project.

The following sources of leakage emissions are accounted for in this tool:

¹⁶ <https://cdm.unfccc.int/methodologies/PAmethodologies/tools/am-tool-14-v2.pdf>

- CH₄ and N₂O emission from composting of digestate;

During the site visit, the assessment team confirmed that the project activity does not include the composting process; therefore, this criterion does not apply to the project.

- CH₄ emissions from the anaerobic decay of digestate disposed in a SWDS or subjected to anaerobic storage, such as in a stabilization pond;

During the site visit, the assessment team confirmed that the digestate from the facility is not sent to a SWDS or kept in a storage or evaporation pond, this criterion does not apply to the project.

- Emission sources associated with N₂O emissions from physical leakages from the digester, transportation of feed material and digestate or any other on-site transportation, piped distribution of the biogas, aerobic treatment of liquid digestate and land application of the digestate are neglected because these are minor emission sources or because they are accounted in the methodologies referring to this tool.

During the site visit, the assessment team confirmed that these emission sources have a minor effect on the project activity.

3.3.3 Project Boundary

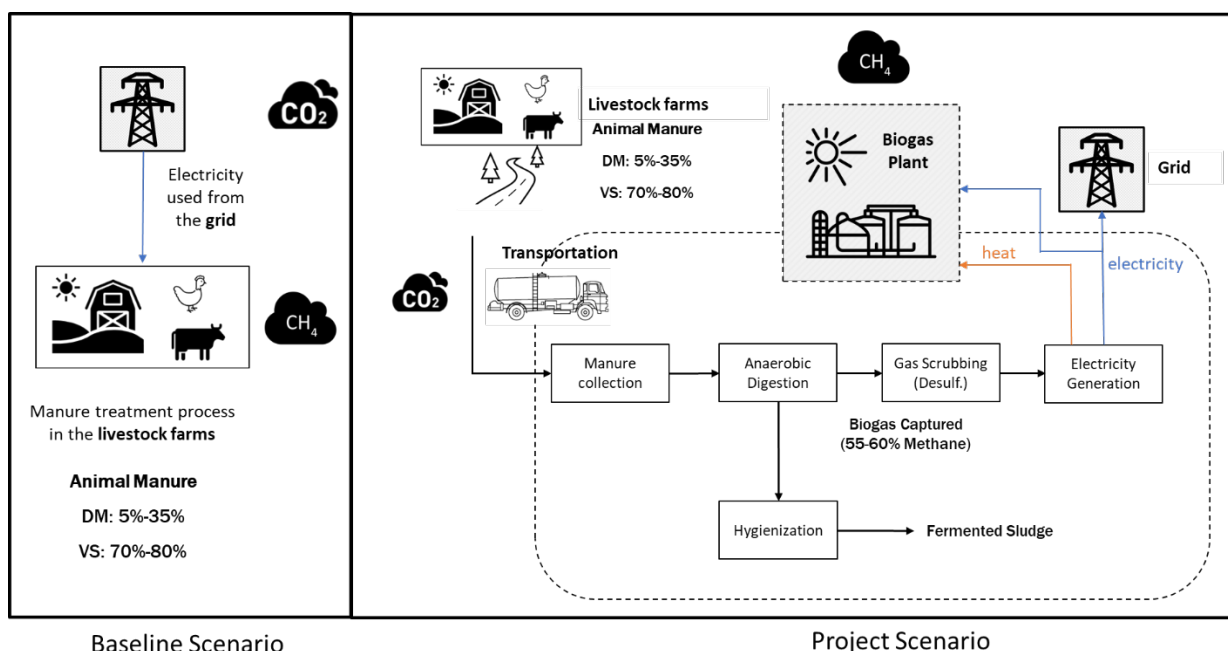
Project boundary has been ascertained and confirmed during the interview using AMS III.D version 21.0 - "Methane recovery in animal manure management systems" and AMS-I.D. "Grid connected renewable electricity generation", version 18.0

The PP defines the project boundary as below which is in accordance with the applied methodologies.

- The central treatment plant;
- The livestock farms where the waste is generated and disposed in the baseline;
- The site of the biogas combustion or energy generation facility;
- The manure storage tanks;
- The road itineraries and/or piping system between the manure collection points and the central treatment plant;

Baseline scenario - The boundaries of the baseline scenario are onsite emissions occurring because current manure management do not mitigate GHG emissions.

Project activity scenario - The boundaries of the project activity scenario are onsite emissions occurring from treatment process.



The geo co-ordinates of the same is discussed in the VCS-PD, which is cross checked with the google map and confirmed to be in order.

The relevant GHG sources included in or excluded from the project boundary are shown on the Table below.

Source	Gas	Included?	Justification/Explanation
Baseline	CO ₂	No	CO ₂ emissions from the decomposition of organic waste are not accounted
	CH ₄	Yes	The major source of emissions in the baseline
	N ₂ O	No	As per the applied Methodology, this emission is not accounted for.
	Other	-	-
	CO ₂	Yes	Electricity may be consumed from the grid in the baseline scenario
	CH ₄	No	As per the applied Methodology, this emission is not accounted for.
	N ₂ O	No	As per the applied Methodology, this emission is not accounted for.
	Other	-	-
	CO ₂	No	

Source		Gas	Included?	Justification/Explanation
	Emissions from thermal energy generation	CH ₄	No	The VVB confirms that there is no thermal energy generation in the baseline.
		N ₂ O	No	
		Other	-	
Project	Emissions from on-site electricity use	CO ₂	Yes	These emissions are accounted for since the project may consume electricity from the national grid in time of emergent situations.
		CH ₄	No	As per the applied Methodology, this emission is not accounted for.
		N ₂ O	No	As per the applied Methodology, this emission is not accounted for.
		Other	-	-
	Emissions from thermal energy generation	CO ₂	No	The VVB confirms that the project does not utilize thermal energy from fossil fuel sources. Hence, this boundary condition is applicable.
		CH ₄	No	
		N ₂ O	No	
		Other	-	
	Emissions from the manure treatment processes	CO ₂	No	CO ₂ emissions from the decomposition of organic waste are not accounted.
		CH ₄	Yes	The emissions from anaerobic digesters and aerobic treatment are accounted.
		N ₂ O	No	As per the applied Methodology, this emission is not accounted for.
		Other	-	-
	Emissions from manure transportation	CO ₂	Yes	May be an important emission source. Hence, they are accounted.
		CH ₄	No	As per the applied Methodology, this emission is not accounted for.
		N ₂ O	No	As per the applied Methodology, this emission is not accounted for.
		Other	-	-
	Emissions from sludge composting	CO ₂	No	Via site visits the VVB confirms that proposed project activity does not involve composting.
		CH ₄	No	Via site visits the VVB confirms that proposed project activity does not involve composting.

Source	Gas	Included?	Justification/Explanation
	N ₂ O	No	Via site visits the VVB confirms that proposed project activity does not involve composting.
	Other	-	-
	CO ₂	No	CO ₂ emissions from the decomposition of organic waste are not accounted.
	CH ₄	No	The VVB confirms that the manure is not stored more than 24 hours via site visits. Hence, these emissions are not accounted.
	N ₂ O	No	As per the applied Methodology, this emission is not accounted for.
	Other	-	-

By checking the VCS PD and site audit, EcoLance can confirm that all the emission sources, gases have been included in the project boundary, and the description in the VCS PD is accurate and complete, and also that the selected sources and gases are justified for the proposed project activity.

3.3.4 Baseline Scenario

According to applied AMS-III.D Methodology, version 21.0., the baseline scenario is defined as the situation where, in the absence of the project activity, animal manure is left to decay anaerobically within the project boundary and methane is emitted to the atmosphere.

In the pre project scenario animal manure was not supplied to the processing facility, it was left to decay anaerobically, since it was not mandated by any law or regulation. Same has been confirmed with the baseline survey, Turkish regulation on livestock farm¹⁷ and interview with farm owners.

As per CDM Validation and Verification Standard for project activities version 03.0, "where the baseline scenario is not prescribed in the approved methodology, the VVB shall assess the list of identified credible alternatives to the project activity in the CDM PDD selected to determine the most realistic baseline scenario." As the selected small-scale methodology clearly mention the baseline scenario and the same has been opted in this project, therefore, no further analysis on baseline is required.

¹⁷ <http://www.resmigazete.gov.tr/eskiler/2006/08/20060809-4.htm>

Second Component of the Proposed Project Activity as per AMS-I.D., version 18.0

The paragraph 19 of AMS-I.D., version 18.0, defines the baseline scenario for greenfield power plant, like in the case of this project activity, as “the electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources into the grid”.

The validation team confirms that the VCS PD confirms to the guidance given by EB via CDM Validation and Verification Standard for project activities version 03.0 and Verra via VCS Standard version 4.5. Therefore, the validation team confirms that the baseline scenario in the absence of the project activity is as described below:

Methane is released directly into the atmosphere at the livestock farm’s anaerobic manure management system (uncovered anaerobic lagoon) without the use of a methane recovery or destruction system.

3.3.5 Additionality

Additionality conditions are set out at the Paragraph 15 and 16 of the applied methodology AMS-III.D.

Paragraph 15 states that “Project activities may demonstrate the additionality by showing that there is no regulation in the host country, applicable to the project site, that requires the collection and destruction of methane from livestock manure. If so, it is not required to apply the “Guidelines on the demonstration of additionality of small-scale project activities”. Based on the regional and technical expertise of the VVB, it is confirmed that the project activity shall be considered as additional since there is no legal regulation in Turkey that requires the collection and destruction of methane from livestock manure. Same can be confirmed through the Turkish regulation on livestock farm¹⁸.

Hence, the VVB confirms the suitability of this project to the regulatory surplus provisions set out in the VCS Standard version 4.5.

Paragraph 16 states that “This additionality condition also applies to Greenfield project activities. Furthermore, for project activities applying this methodology in combination with a Type I methodology, that has an energy component whose installed capacity is less than 5 MW, this procedure for additionality demonstration also applies to that component.” The

¹⁸ <http://www.resmigazete.gov.tr/eskiler/2006/08/20060809-4.htm>

VVB confirms that the project activity complies with this additionality condition since it is implemented as a greenfield activity with the total capacity of 2.406 MWe. This information is confirmed via site visits and the review of the project feasibility report. Consequently, additionality is justified for the project.

3.3.6 Quantification of GHG Emission Reductions and Removals

The assessment team checked the baseline, project and leakage calculation and confirm that the evaluation of baseline, project, and leakage in line with the approved methodology and formula used to calculate the same is correct. The detail analysis is as below:

Baseline Emissions as per AMS-III.D

In accordance with AMS-III.D “Methane recovery in animal manure management systems”, version 21.0, baseline emissions (BE_y) shall be calculated by utilizing one of the two options as given below:

- (a) Using the amount of the waste or raw material that would decay anaerobically in the absence of the project activity, with the most recent IPCC Tier 2 approach (please refer to the chapter ‘Emissions from Livestock and Manure Management’ under the volume ‘Agriculture, Forestry and other Land use’ of the 2006 IPCC Guidelines for National Greenhouse Gas Inventories). For this calculation, information about the characteristics of the manure and of the management systems in the baseline is required. Manure characteristics include the amount of volatile solids (VS) produced by the livestock and the maximum amount of methane that can be potentially produced from that manure (B_0);
- (b) Using the amount of manure that would decay anaerobically in the absence of the project activity based on direct measurement of the quantity of manure treated together with its specific volatile solids (SVS) content.

Option 17 (b) has been selected for the calculation of baseline emissions by the PP. As per AMS-III.D, baseline emissions are calculated as given below:

$$BE_y = GWP_{CH_4} \times D_{CH_4} \times UF_b \times \sum_{j,LT} (MCF_j \times B_{0,LT} \times Q_{manure,j,LT,y} \times SVS_{j,LT,y})$$

Where:

BE_y	= Total baseline emissions in year y, in tCO ₂ e/year
$GW P_{CH_4}$	= Global Warming Potential of CH ₄ applicable to the crediting period (28 tCO ₂ e/tCH ₄)
D_{CH_4}	= CH ₄ density (0.00067 t/m ³ at room temperature (20°C) and 1 atm pressure)
UF_b	= Model correction factor to account for model uncertainties (0.94)
LT	= Index for all types of livestock
j	= Index for animal manure management system
MCF_j	= Annual methane conversion factor for the baseline animal manure management system j
$B_{0,LT}$	= Maximum methane producing potential of the volatile solid generated for animal type LT (m ³ CH ₄ /kg-dm)
$Q_{manure,j,LT,y}$	= Quantity of manure treated from livestock type LT and animal manure management system j (tonnes/year, dry basis)
$SVS_{j,LT,y}$	= Specific volatile solids content of animal manure from livestock type LT and animal manure management system j in year y (tonnes/tonnes, dry basis)

Estimation of $B_{0,LT}$ and MCF_j :

As per AMS-III.D methodology, $B_{0,LT}$ can be determined by using default values from IPCC Guidelines, as updated in July 2023 and direct measurement if there are no country specific data available. The PPP has chosen to use the default data for this project activity which is suitable as there are no available country specific data as per the VVBs regional expertise. The default values have been selected to determine $B_{0,LT}$ values and these values are obtained from 2019 IPCC Refinement, as updated in July 2023 Table 10.16A (updated), (p. 10.67, Other Regions, High Productivity Systems). The VVB confirms that Turkey falls into the category of other regions as per the IPCC Guideline and the animal farms that provides manure to the project are considered as high productivity systems.

As per AMS-III.D methodology, MCF_j values shall be obtained from 2019 Refinement to the 2006 IPCC Guideline, as updated in July 2023 is there are no country specific data available. The VVB confirms that using default IPCC values is suitable since there is no country specific data available for Turkey. In accordance with IPCC Guideline, MCF value shall be determined based on climate zone which is determined in relation to mean annual temperature, mean annual precipitation, and evapotranspiration. Following the procedure to determine the climate zone in 2019 Refinement to the 2006 IPCC Guideline Volume 4 Chapter 10 Table 10.17 (Updated) (page 10.68). The VVB confirms that the project falls into the warm temperature dry climate zone upon checking the provided

mean annual temperature, mean annual precipitation, and evapotranspiration data by the PP.

The fixed parameters used to calculate BE_y is determined and verified as given below:

Parameter	Value	Unit
GWP _{CH4}	28	Dimensionless
ρ _{CH4}	0.000	t/m ³
f _{CH4,default}	0.6	Nm ³ CH ₄ /Nm ³ biogas
MCF _j	0.76	Fraction
B _{0,Other} Cattle	0.18	m ³ CH ₄ / kg-dm
UF _b	0.94	Fraction

PP has used default values for baseline emissions as per AMS-III.D which is consistent with the IPCC guidelines, as updated in July 2023. The validation team has checked the default values used for baseline calculations and confirms the consistency with the Guidelines.

BE_y=30,188 tCO_{2e}/yr

Baseline Emissions as per AMS.I-D

In accordance with the paragraph 22 in the applied methodology, baseline emissions include only CO₂ emissions from electricity generation in power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are to be calculated as follows:

$$BE_y = BE_{EC,y} = EG_{PJ,y} \times EF_{grid,y}$$

Where:

BE_y = Baseline emissions in year y (t CO₂)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh)

$EF_{grid,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (t CO₂/MWh)

Since the proposed project activity is the installation of a greenfield power plant, as per the paragraph 26 of AMS-I.D, version 18., $EG_{PJ,y}$ is calculated as follows:

$$EG_{PJ,y} = EG_{PJ,facility,y}$$

Where:

$EG_{PJ,facility,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in year y

As per the paragraph 23 of AMS-I.D, version 18., the emission factor shall be calculated in a transparent and conservative manner. The project participants choose to calculate a combined margin (CM), consisting of the combination of operating margin (OM) and build margin (BM) according to the procedures prescribed in the “Tool to calculate the emission factor for an electricity system”, version 07.0.

Based on 2020 data, Ministry of Energy and Natural resources of Turkey¹⁹ has calculated combined margin and published the data:

$$EF_{grid,CM,y} = 0.5552 \text{ tCO}_2/\text{KWh}$$

¹⁹

<https://enerji.gov.tr//Media/Dizin/EVCED/tr/%C3%87evreVe%C4%B0klim/%C4%B0klimDe%C4%9Fi%C5%9Fikli%C4%9Fi/TUES EmisyonFktr/Belgeler/Bform2020.pdf>

The validation team has checked the emission factor calculations, as mapped out in the PD, and confirms that the calculations are in line with the application of the project activity and the country-specific data that has been published by the Turkish Government.

$$BE_y = BE_{EC,y} = EG_{PJ,y} \times EF_{grid,y}$$

$$BE_{EC,y} = 6,658 \times 0.5552$$

$$BE_{EC,y} = 3,696 \text{ tCO}_2\text{e/yr}$$

Project emissions as per AMS.III-D

As per AMS-I.D, project activity emissions consist of:

$$PE_y = PE_{PL,y} + PE_{flare,y} + PE_{power,y} + PE_{transp,y} + PE_{storage,y}$$

Where:

PE_y = Project emissions in year y (t CO₂e)

$PE_{PL,y}$ = Emissions due to physical leakage of biogas in year y (t CO₂e)

$PE_{flare,y}$ = Emissions from flaring or combustion of the biogas stream in the year y (t CO₂e)

$PE_{power,y}$ = Emissions from the use of fossil fuel or electricity for the operation of the installed facilities in the year y (t CO₂e)

$PE_{transp,y}$ = Emissions from incremental transportation in the year y (t CO₂e), as per relevant paragraph in AMS-III.AO

$PE_{storage,y}$ = Emissions from the storage of manure (t CO₂e)

Project emissions due to physical leakage of biogas ($PE_{PL,y}$):

As per the paragraph 21(b) of AMS-III.D., version 21.0., project emissions due to physical leakage of biogas from the animal manure management systems used to produce, collect, and transport the biogas to the point of flaring or gainful use are determined as, in accordance with the paragraph 23 of the methodological tool “Project and leakage emissions from anaerobic digesters, version 02.0. According to the methodological tool, there are two different procedures to determine the quantity of methane produced in the digester in year y ($Q_{CH4,y}$).

The PP has chosen to apply Option 2 in the applied “Tool to calculate the quantity of methane produced in the digester”.

$$Q_{CH4,y} = Q_{biogas,y} \times f_{CH4,default} \times \rho_{CH4}$$

$$PE_{PL,y} = PE_{CH_4,y} = Q_{CH_4} \times EF_{CH_4,y} \times GWP_{CH_4}$$

The parameter $EF_{CH_4,default}$ has been used as 0.028 according to the applied tool “Project and leakage emissions from anaerobic digesters, version 02.0. The default factor for digesters with steel or lined concrete or fiberglass digesters and a gas holding system (egg shaped digesters) and monolithic construction has been considered. Upon site visit, the validation team has checked the default emission factor used for the fraction of CH₄ produced that leak from the anaerobic digester and confirms the consistency with the type of digester installed at the plant.

The parameter $f_{CH_4,default}$ has been used as 0.6 in accordance with the applied tool Project and leakage emissions from anaerobic digesters, version 02.0. Since the PP has chosen Option 2 for determination of the quantity of methane produced in the digester, the parameter of $f_{CH_4,default}$ is used as 0.6 in the related calculation. The validation team has checked default value for the fraction of methane in the biogas in the applied tool and confirmed the related parameter as 0.6.

Project emissions from flaring or combustion of the biogas stream ($PE_{flare,y}$):

With regards to project emissions from flaring, the applied Methodology AMS-III.D states that “In the case of flaring of the recovered biogas, project emissions are estimated using the procedures described in the methodological tool “Project emissions from flaring”. If the recovered biogas is combusted for electrical/thermal energy production or for other gainful use, the methane destruction efficiency can be considered as 100%. However, this use of the recovered biogas shall be included in the project boundary and its output shall be monitored in order to ensure that the recovered biogas is actually destroyed, even if the emission reductions from this component are not claimed.” Upon site visits and review of official project documents such as the EIA report and electricity generation license, the VVB confirms that the biogas is utilized for electricity generation.

Electricity generation is included in the project boundary accordingly. Biogas flow and electricity output from the gas engines is monitored inline with the applied Methodology.

An emergency flare is installed within the project activity to ensure that biogas would be destroyed when exigencies happened. The flare will only be used when the gas engine is not in operation. As confirmed with PP, emission reductions from destroying methane by flare will not be claimed during the crediting period, which is conservative.

Since flare is installed for only emergency purposes and the main purpose of the recovered biogas is electricity generation, the flare efficiency shall be considered as 100% and $PE_{flare,y} = 0$.

In case of biogas flaring, the project emission shall be calculated using the equation below;

$$PE_{flare,y} = GWP_{CH4} \times \sum_{m=1}^{525600} F_{CH4,RG,m} \times (1 - \eta_{flare,m}) \times 10^{-3}$$

$PE_{flare,y}$ = Project emissions from flaring of the residual gas in year y (tCO₂e)

GWP_{CH4} = Global warming potential of methane valid for the commitment period (tCO₂e/tCH₄)

$F_{CH4,RG,m}$ = Mass flow of methane in the residual gas in the minute m (kg)

$\eta_{flare,m}$ = Flare efficiency in the minute m

Project emissions from electricity and fossil fuel consumption ($PE_{power,y}$):

Since the electricity requirement of the project is supplied with the electricity obtained from the biogas production, there is no electricity consumption from the national grid in the project activity and $PE_{EC,y}$ has been considered as zero by the PP. Upon site visit, the validation team confirms that the project activity supplies electricity consumption associated with the anaerobic digestion plant only from on-site electricity generation. However, if project consumes electricity from the national grid due to any circumstance, such as emergency, the relevant calculations will be included. The VVB confirms that the relevant equations are provided in the VCS-PD and the monitoring equipment for possible electricity consumption from the grid is in place.

Project emissions from incremental transportation ($PE_{transp,y}$):

In accordance with the AMS-III.D, version 21.0, $PE_{trans,y}$ is calculated in line with the paragraph 14 of AMS-III.AO, version 01.0, project emissions from incremental transport distances are calculated depending on the incremental distances between.

$$PE_{trans,y} = (Q_y/CT_y) \times DAF_w \times EF_{CO2,transport}$$

Where:

Q_y = Quantity of raw waste/manure treated and/or wastewater co-digested in the year y (tonnes)

CT_y = Average truck capacity for transportation (tonnes/truck)

DAF_w = Average incremental distance for raw solid waste/manure and/or wastewater transportation (km/truck)

$EF_{CO_2,transport}$ = CO_2 emission factor from fuel use due to transportation ($kgCO_2/km$, IPCC default values or local values may be used)

As per the methodological tool “Project and leakage emissions from road transportation of freight” Version 01.0.0, it is stated as below:

“Vehicle class. This tool defines two vehicle classes based on their gross vehicle mass:

- Light vehicles. Vehicles with a GVM being less or equal to 26 tonnes;
- Heavy vehicles. Vehicles with a GVM being higher than 26 tonnes”

In addition to this, $EF_{CO_2,transport}$ has also been determined according to the class of vehicle used in the project activity. Values of this parameter are used as $0.245\text{ kgCO}_2/km$ for light vehicles and $129\text{ kgCO}_2/km$ for heavy vehicles, respectively.

In line with the applied methodological tool, CO_2 emission factor from fuel due to transportation has been determined as $0.245\text{ kgCO}_2/km$ by the PP according to class of vehicle, which is used in the project activity. Based on vehicle licenses and google map data showing between distance biogas plant and animal farms, the validation team confirms that light vehicles have been used within the project activity and checked that the average distance is $14,3\text{ km/truck}$. Hence, the applied values are suitable.

Project emissions from storage of manure ($PE_{storage,y}$):

As per the paragraph 24 of AMS-III.D., version 21.0., project emissions on account of storage of manure before being fed into the anaerobic digester shall be accounted for if both condition (a) and condition (b) below are satisfied:

- (a) The storage time of the manure after removal from the animal barns, including transportation, exceeds 24 hours before being fed into the anaerobic digester;
- (b) The dry matter content of the manure when removed from the animal barns is less than 20%.

The PP has considered the projects emissions from storage of manure as zero since the manure supplied from animal farms is not stored in outdoor open storage tanks for more than 24 hours at the plant. Based on the baseline survey and interviews, the validation

confirms that project emissions from manure storage can be omitted that manure is not stored in outdoor open storage tanks for more than 24 hours.

Leakage emissions as per AMS.III-D

As per the applied methodology, leakage is determined by following the relevant procedure in the methodological tool “Project and leakage emissions from anaerobic digesters”. The tool accounts for leakage emissions associated with storage and composting of digestate.

Within the scope of the project activity, digestate obtained from biogas production is distributed free of charge to the neighboring cropland and farm owners without any storage under anaerobic conditions or composting process. In other words, in Section 2.2 of the methodological tool “Project and leakage emissions from anaerobic digesters”, it is stated as follows:

- “ (a) CH₄ and N₂O emission from composting of digestate;
- (b) CH₄ emissions from the anaerobic decay of digestate disposed in a SWDS or subjected to anaerobic storage, such as in a stabilization pond”.

Considering the methodological tool, the project activity does not involve composting of the digestate or the anaerobic decay of digestate disposed in a SWDS or subjected to anaerobic storage, such as in a stabilization pond. Therefore, leakage emissions associated with the anaerobic digesters are not taken into account for the project activity. Same has been confirmed during the on-site audit.

The fermented product has been distributed to neighboring farms, there is no storage of fermented product under anaerobic conditions and no composting process. Therefore, the PP has not taken into account emissions from leakage. The protocol signed with the cropland and farm owners where the fermented product was applied has been checked by the validation team. Upon the related documents and interviews with the cropland and farm owners, the validation team confirms that both the storage of digestate within the project boundary and no composting is applied to the project activity. Hence, exclusion of leakage emissions is confirmed to be inline with the applied Methodology and the applied Tool14.

Net GHG Emissions Reductions and Removals:

$$ER_y = BE_y - PE_y$$

Where:

ER_y = Emission reductions in year y (tCO₂e/yr)

BE_y = Baseline emission in year y (tCO₂e/yr)

PE_y = Project emission in year y (tCO₂e/yr)

As per para 27 of the applied methodology ex-post emissions shall be determined through direct measurement of the amount of methane fuelled, flared or gainfully used following the equation below:

$$ER_{y,ex-post} = \min[(BE_{y,ex-post} - PE_{y,ex-post}), (MD_y - PE_{power,y,ex-post})]$$

Where;

$ER_{y,ex-post}$ = Emission reductions achieved by the project activity based on monitored values for year y (t CO₂e)

$BE_{y,ex-post}$ = Baseline emissions calculated using equation 1

$PE_{y,ex-post}$ = Project emissions calculated using equation 6 using ex post monitored Values

MD_y = Methane captured and destroyed or used gainfully by the project activity in year y (t CO₂e)

$PE_{power,y,ex-post}$ = Emissions from the use of fossil fuel or electricity for the operation of the installed facilities based on monitored values in the year y (t CO₂e)

For projects that use captured methane for electricity generation, MD_y shall be calculated using the equation below based on the amount of monitored electricity generation:

$$MD_y = \frac{EG_y}{NCV_{CH_4} * EE_y} * D_{CH_4} * GWP_{CH_4}$$

Where;

EG_y = Total electricity generated from the recovered biogas in year y (MWh)

3600 = Conversion factor (1 MWh = 3600 MJ)

NCV_{CH_4} = NCV of methane (MJ/Nm³) (use default value: 35.9 MJ/Nm³)

EE_y = Energy conversion efficiency of the project equipment, which is determined by adopting one of the following criteria:

- Specification provided by the equipment manufacture. The equipment shall be designed to utilize biogas as fuel, and efficiency specification is for this fuel. If the specification provides a range of efficiency values, the highest value of the range shall be used for the calculation;

- Default efficiency of 40 %.

As per the EIA report and electricity generation license, the biogas generated by anaerobic digestion is utilized for electricity generation through gas engines. During the sites visits, it is confirmed that the amount of generated electricity is continuously measured by electricity meters installed at the facility. Therefore, it is confirmed only methane combusted in the gas engines would be counted for ER calculation.

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
31/12/2020	93	3	0	90
01/01/2021 - 31/12/2021	33,884	1,027	0	32,857
01/01/2022 - 31/12/2022	33,884	1,027	0	32,857
01/01/2023 - 31/12/2023	33,884	1,027	0	32,857
01/01/2024 - 31/12/2024	33,884	1,027	0	32,857
01/01/2025 - 31/12/2025	33,884	1,027	0	32,857
01/01/2026 - 31/12/2026	33,884	1,027	0	32,857
01/01/2027 - 30/12/2027	33,791	1,024	0	32,767
Total	237,188	7,189	0	229,999

In conclusion, the VCS-PD includes all the parameters that are applied in the estimation of emission reductions.

There are no uncertainties associated with the calculation of emissions.

All estimates of the emission reductions can be replicated using the data and parameter values provided in the project description and complying with the provisions of the applied methodology.

The methodology and any referenced tools have been applied correctly to calculate baseline emissions, project emissions, leakage and net GHG emission reductions and removals. The entire emission reduction calculations have been done in separate worksheets that was reviewed by the validation team and found meeting the requirements.

3.3.7 Methodology Deviations

No methodology deviation has been sought.

3.3.8 Monitoring Plan

The validation team evaluated the monitoring plan through site visits and interviews with the project manager and staff and confirms that the monitoring plan has been constructed in line with the approved baseline and monitoring methodology “AMS-III.D”, version 21.0 and “AMS-I.D”, version 18.0.

For the evaluation, the validation team has crosschecked the provided information on monitoring equipment with supporting documents, such as calibration details, and confirmed the suitability of the monitoring equipment to the monitoring plan. Upon site visits and document review, it is confirmed that the monitoring plan will give opportunity for real measurement of achieved emission reductions. EcoLance’s team has checked all the parameters presented in the monitoring plan against the requirements of the methodology; no deviations relevant to the project activity have been found in the plan.

EcoLance team confirms that the monitoring arrangements described in the monitoring plan are feasible within the project design, and the means of implementation of the monitoring plan are sufficient to ensure the emission reductions achieved by/resulting from the proposed VCS project activity can be reported ex post and verified.

Ex – ante parameters for the project activity:

Parameters	VVB Assessment
GWP_{CH_4} - Global Warming Potential for CH ₄	PP has used IPCC Fifth Assessment Report, Chapter 8, Table 8.A.1 value as the source of document which has been cross verified by the assessment team and found consistent. Value – 28 tCO₂e/tCH₄
D_{CH_4} - Density of methane at normal (at room temperature 20°C and 1 atm pressure) conditions	PP has applied CDM approved methodology value as a source of document which has been checked by the assessment team and found consistent. Value – 0.00067 t/m³
$f_{CH_4, default}$ – Default value for the fraction of methane in the biogas	PP has applied the default value from Tool14: Project and leakage emissions from anaerobic digesters as per the applied CDM approved methodology. The VVB confirms that the use of Option 2 at the applied Tool 14, which allows for the use of default value for the fraction of methane in the biogas, is suitable. Value – 0.6
UF_b – Model correction factor	PP has applied the value given as in the applied CDM approved methodology which has been cross by the assessment team and found consistent. Value – 0.94
$EF_{CH_4, default}$ – Default emission factor for the fraction of CH ₄ produced that leak from the anaerobic digester	According to applied Tool 14, default emission factor for the fraction of CH₄ that leaks from the anaerobic digester shall be determined for the specific digester type. The VVB confirms that the type of digesters chosen for this default value (digesters with steel or lined concrete or fiberglass digesters and a gas holding system and

	monolithic construction) is consistent with the project activity. Value – 0.028 t CH₄ leaked/ t CH₄ produced
$EF_{CO_2,transport}$ - CO₂ emission factor from fuel use due to transportation	PP has used 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 2 value as the source of document which has been cross verified by the assessment team and found consistent. Value – 0.245 kgCO₂/km

Ex – post parameters for the project activity:

Parameters	VVB Assessment
$Q_{manure,LT,y}$ - Quantity of manure treated from livestock type LT at animal manure management system j	VVB has checked the weighbridge records furnished that gives this value as wet basis, along with the dry matter measurements carried out in the lab for each manure received from the farms. Upon checking the weighbridge / lab logbook records, along with confirming the calibration of the associated equipment complies with the national regulations, the assessment team has validated this parameter. Value – Please refer VCS PD for the same
Q_y – Quantity of raw manure co-digested in the year y	VVB has reviewed the weighbridge log book records as well as the calibration details of the weighbridge, and validated the consistency of the parameter. Value – Please refer VCS PD for the same
$SVS_{LT,y}$ – Specific volatile solids content of animal manure from livestock type LT and animal manure management system j in year y	The validation team has confirmed that the parameter is calculated by the lab staff in line with the Annex 2 of the methodology AM0073. Upon checking the lab logbook records along with confirming the calibration of the associated lab equipment complies with the national regulations, the

	assessment team validated this parameter. The sampling plan described in Section 5.2 and 5.3 of the VCS-PD is confirmed to be in line with the methods described in the applied Methodology and consistent with the laboratory applications. Value – Please refer VCS PD for the same
$B_{0,LT}$ - Maximum methane producing potential of the volatile solid generated for animal type 'LT'	PP has used IPCC 2019, as updated in July 2023 value as a source of document & which has been checked by the assessment team and found consistent. Please refer to Section 3.3.6 for further assessment on the selection of this value. Value Non-Dairy Cattle: 0.18 m³CH₄/kg-VS-dm
CT_y – Average truck capacity for transportation	PP has used project proponent records of the net capacity of the trucks at weighbridge reading as a source of document & which has been cross verified by the assessment team upon checking the vehicle licenses and found consistent. Value – 21 tonnes/truck
DAF_w – Average incremental distance for raw solid manure transportation	The PP measurements of distance between the central treatment plant and the manure supplying farms as documented in logbooks are cross verified upon checking the geo-coordinates of the farms. The locations have been validated as well during the site visit. Value – 14.3 km/truck
$EG_{PJ,y} = EG_{facility,y}$ - Quantity of net electricity generation supplied by the project plant/unit to the grid in year y	This parameter is continuously monitored through electricity meters and cross-checked by using electricity invoices /project proponent records as a source of document which has been cross verified by

	the assessment team and found consistent. The calibration of the electricity meters has also been found in compliance with the national regulations. Value –6,658 MWh/yr
$EC_{PJ,j,y}$ - Quantity of electricity consumed by the project electricity consumption source the national grid in year y	The electricity requirement of the project will be supplied with the electricity obtained from the biogas production. Therefore, there will be no electricity consumption from the grid, this value was considered as zero. The assessment team validates this by checking the reading from the electricity meters. Value –0
$EF_{EF,j,y} = EF_{grid,y}$ - Emission factor for the Turkish National Grid	PP has used the Ministry of the Energy and Natural Resources in Turkey records as a source of document which has been cross verified by the assessment team and found consistent and up-to-date. Value – 0.5552 tCO₂e/MWh
$TDL_{j,y}$ - Average technical transmission and distribution losses for providing electricity to source j in year y	PP has used data from Turkish Electricity Transmission Corporation (TEİAŞ) as a source of document which has been cross verified by the assessment team upon checking the given references and found consistent. Value – 0.11
$Q_{biogas,y}$ – Amount of biogas collected at the digester outlet in year y	PP measured this value instantaneously by the flowmeter installed at the collector gas line leaving the digesters and aggregated to determine monthly and yearly biogas amount. Besides confirming the calibration of flowmeters in compliance with the national regulations. For ex-ante estimation of this parameter, the PP has used one-year SCADA export which the

	assessment team has checked and confirmed. Value – 3,197,940 Nm³ biogas (Please refer to PD Section 5.2 for detailed justification)
Q _{CH₄,y} - Quantity of methane produced in the anaerobic digester in year y	This parameter shall be calculated with the values monitored continuously by the flowmeter installed at the collector gas line the calibration details of which has been checked and validated by the VVB. The obtained data is archived electronically. The PP calculates this value by multiplying with the methane content of the biogas (0.6) which is found to be consistent with the applied methodology. Value – 1,286 t CH₄/y
MCF _j - Annual methane conversion factor (MCF) for the baseline animal manure management system j	This value shall be determined based on the IPCC 2019 guidelines, as updated in July 2023 according to which the climate zone of the project activity shall be monitored. The VVB has confirmed the climate zone of the project activity as Warm Temperature Dry. Any change on IPCC Guidelines as well as the meteorological data from the project location must be monitored for the parameter. Value – 0.76
DM - Residue remaining after water is removed from waste material by evaporation; dry matter	The VVB has checked the project proponent records and lab records and found the reported values consistent. Further, the related monitoring laboratory equipment and the applied monitoring measurement methods have been checked during the site visits and found to be suitable to the applied methodology. Value – Please refer VCS PD for the same

$\eta_{flare,m}$ - The flare efficiency	The flare efficiency is determined in accordance with the Paragraph 22 of the applied Methodology which is confirmed to be suitable by the validation team. The VVB also confirms that necessary monitoring equipment is installed at the plant for the monitoring of flare efficiency in case any biogas is flared. Value – 100%
EE _y - Energy conversion efficiency of the project equipment	Default efficiency of 40% is used in line with the applied Methodology para 30. Value – 40%
EG _y - Total electricity generated from the recovered biogas in year y (MWh)	EG_y is monitored continuously by electricity meters. The monitored values are recorded on a daily basis, aggregated monthly and yearly. The electricity meters are calibrated in accordance with national calibration standards.

The monitoring plan as included in the VCS-PD covers all the relevant monitoring parameter as required in the context of the project activity and as prescribed in the applied methodology and tools. The monitoring systems and responsibilities are clearly defined in the VCS-PD and the validation team confirms that the monitoring plan is feasible and implementable in the context of the project design. The monitoring parameters meet the minimum requirement of the applied methodology/tools in terms of measurement method, frequency and associated quality checks/ cross checks. The validation team has checked and confirmed the suitability of each monitoring equipment and the calibration details related to each. The procedures for internal auditing and QA/QC are separately defined for each parameter, wherever applicable.

The sampling on SVS_{LT,y} parameter has been evaluated and found appropriate and in line with the applied Methodology. The monitoring plan has not applied sampling on any parameter other than SVS_{LT,y}, and all of these are directly monitored or estimated based on monitoring methodology prescribed in applied methodology and tools, as appropriate. The organizational structure, responsibilities and competencies of the personnel that will be carrying out

monitoring activities are adequately defined in the VCS PD including the policies for oversight and accountability of monitoring activities. Relevant internal procedures identified for day-to-day records handling (including what records to keep, storage area of records and how to process performance documentation) are suitable for monitoring.

It has been included in the VCS PD that all monitored data required for verification and issuance will be kept for five years after the end of the crediting period or the last issuance of VCUs, for this project activity, whichever occurs later.

3.4 Non-Permanence Risk Analysis

Not applicable.

4 VALIDATION OPINION

EcoLance has performed a validation of the project activity “Elmadag Biogas Power Plant” with VCS ID-3890. The validation was performed on the basis of VCS Version 4.5 requirements as well as criteria given to provide for consistent project operations, monitoring and reporting.

The responsibility of the project proponent is the GHG statement. Based on the review of the project design documentation and the subsequent follow-up interviews provided, the assessment team determined that the project conforms with the validation criteria for projects set out in VCS Version 4.5.

The project correctly applied the following baseline and monitoring methodologies.

AMS.III-D “Small-scale methodology – Methane recovery in animal manure management systems”, Version 21.0.

AMS.I-D “Grid connected renewable electricity generation”, Version 18.0.

The purpose of the proposed project activity is to generate renewable energy by capturing biogas from animal manure -via anaerobic digestion- and utilizing it to produce electric energy through gas engines. The project enables reduction of GHG incurred from the previous manure management system, which was uncovered anaerobic lagoon.

This project adopts the renewable crediting period of 7 years, twice renewable for a total of 21 years. The estimated average net annual emission reductions are 32,857 tCO₂eq/y.

Accordingly, the project is expected to generate 229,999 tCO₂eq emissions reduction during the first crediting period.

The emission reduction forecast has been checked and it is deemed likely that the stated amount is achieved given that the underlying assumptions do not change. The main historical assumption is the default IPCC values used during the quantification of ER, and the main projected assumption is the status of the animal farms considered in the baseline emissions. During the monitoring period, potentially new manure suppliers might be added or some of the current ones might be excluded, and as well, the animal number or type can change. The actual results may be subject to change in case of any deviation in these assumptions.

The monitoring plan sufficiently describes the monitoring of the project's emission reductions. The monitoring arrangements described in the monitoring plan are feasible within the project design and it is EcoLance's opinion that the project participants are able to implement the monitoring plan.

In summary, it is EcoLance opinion that the project activity "Elmadag Biogas Power Plant" as described in the PD version 3 dated 08/12/2023 meets all relevant VCS Standard version 4.5 requirements and correctly applies the small-scale methodology AMS-III.D Methane recovery in animal manure management systems", Version 21.0 and AMS-I.D.: "Grid connected renewable electricity generation", Version 18.0²⁰

Estimated emission reduction for the crediting period is as follows:

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
31/12/2020	93	3	0	90
01/01/2021-31/12/2021	33,884	1,027	0	32,857
01/01/2022-31/12/2022	33,884	1,027	0	32,857

²⁰ <https://cdm.unfccc.int/methodologies/DB/W3TINZ7KKWCK7L8WTXFQQOFQQH4SBK>

01/01/2023-31/12/2023	33,884	1,027	0	32,857
01/01/2024-31/12/2024	33,884	1,027	0	32,857
01/01/2025-31/12/2025	33,884	1,027	0	32,857
01/01/2026-31/12/2026	33,884	1,027	0	32,857
01/01/2027-30/12/2027	33,791	1,024	0	32,767
Total	237,188	7,189	0	229,999

APPENDIX 1: DOCUMENTS REVIEWED

No	Author	Document Title	References to the Document
1	ALMER Çevre Denetim Müşavirlik Mühendislik İş Sağlığı ve Güvenliği Proje Tic. Ltd. Şti.	Detailed Feasibility Report of Elmadağ Project	28.11.2016
2	Energy Market Regulatory Authority (EMRA), Turkey	Electricity Generation Licence	1.08.2019
3	Energy Markets Corporation (EPIAŞ)	Electricity Generation Records	2020 – 2021 – 2022
4	IPCC	2006 IPCC guidelines for National Greenhouse Gas inventories, Volume 2	2006

5	IPCC	2019 Refinement to 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Volume 4, as updated in July 2023	2019
6	IPCC	Fifth Assessment Report, Synthesis Report	2018
7	Manufacturer	Muffle Furnace, NASTECH Series, Technical Specification	-
8	Manufacturer	Dry Air Sterilizer, NÜKLEON NST Series, Technical Specification	-
9	Manufacturer	Analytical Balance, OHAUS PR Series, Technical Data Sheet	-
10	Manufacturer	Moisture Content Analyzer, SHIMADZU MOC63u, Technical Specifications	-
11	Manufacturer	Gas engine supply contract	26.10.2020
12	Manufacturer	Calibration test certificates for main and backup electricity meters MAKEL C250.AMT.2556	05/10/2020 – 10/10/2020
13	Manufacturer	Technical Details and calibration test certificates for biogas flow meter, Endress+Hauser	26.08.2020
14	Manufacturer	Technical Details and calibration test certificate for biogas flow meter, Endress+Hauser	31.08.2020
15	Manufacturer	Weighbridge Invoice, technical details, and calibration test certificate	20.10.2020
16	Ministry of Energy and Natural Resources of Turkey	Project Commissioning Date Certificate	31.12.2020
17	Ministry of Energy and Natural Resources of Turkey	Emission Factor for the Turkish National Grid	20.09.2022

18	Ministry Of Environment, Urbanization and Climate Change	EIA approval report	1.03.2017
19	Ministry of Environment, Urbanization and Climate Change, Turkey	Environmental Impact Assessment Approval	1.03.2017
20	PP	Emission Reduction spread sheets version 3	08.12.2023
21	PP	Central Coordinates of the Project	
22	PP	Baseline Surveys Conducted with Animal Farm Owners	2020 – 2021 – 2022
23	PP	Licenses of the manure/treated effluent carrying trucks	-
24	PP	Monthly Records on Electricity Meters	-
25	PP	Photos of the plant site, laboratory, meters, animal farms, lagoons at farms	2020 – 2021 – 2022
26	PP	GPS coordinates of animal farms supplying manure to the project	-
27	PP	Manure supply agreements with animal farms	2020 – 2021 – 2022
28	PP	Electricity bills raised for the electricity supply to the National Grid	2020 – 2021 – 2022
29	PP	Flare Invoice	20.10.2020
30	PP	Operational Logbook Records	2020-2021-2022
31	PP	Yearly employment records	2021 – 2022
32	PP	Occupational Health and Safety Training Records	2020 – 2021 – 2022

33	PP	VCS PD for “Elmadag Biogas Power Plant” version 3	08.12.2023
34	PP	Meteorological Data: https://mgm.gov.tr/veridegerlendirme/il-ve-ilceler-istatistik.aspx?m=ANKARA	5.12.2020
35	Republic of Turkey Energy Market Regulatory Authority (YEKDEM)	Final List on Renewable Energy Plants	2022
36	Turkish Electricity Transmission Corporation (TEİAŞ)	Average technical transmission and distribution losses	-
37	Turkish State Meteorological Service Directory	Mean Annual Temperature for the Project’s Location, Ankara, Turkey	-
38	Turkish State Meteorological Service Directory	Mean Annual Precipitation for the Project’s Location, Ankara, Turkey	-
39	Turkish Official Gazette	The Measurement and Measuring Instrument Regulations (mevzuat.gov.tr) ²¹	
40	UNFCCC/CDM	AMS-III.D “Methane recovery in animal manure management systems”, Version 21.0	-
41	UNFCCC/CDM	AMS-I.D “Grid connected renewable electricity generation”, Version 18.0	-
42	UNFCCC/CDM	Tool to calculate project or leakage CO2 emissions from fossil fuel combustion, version 03.0	-

²¹<https://www.mevzuat.gov.tr/anasayfa/MevzuatFihristDetayIframe?MevzuatTur=7&MevzuatNo=6381&MevzuatTertip=5#:~:text=Naklimetre%20ve%20hububat%20muayene%20aletlerinin,ve%20Ayar%20Te%C5%9Fkilat%C4%B1%20taraf%C4%B1ndan%20yap%C4%B1%C4%B1r>

43	UNFCCC/CDM	Baseline, project and/ or leakage emissions from electricity consumption and monitoring of electricity generation, version 03.0	-
44	UNFCCC/CDM	Project emissions from flaring, version 04.0	-
45	UNFCCC/CDM	Methodological tool: Project and leakage emissions from anaerobic digesters, Version 02.0	-
46	UNFCCC/CDM	Tool to calculate the emission factor for an electricity system, version 07.0	-
47	UNFCCC/CDM	Validation and Verification standard for project activities, version 03.0 of 09/09/2021 & 11/03/2022	-
48	VERRA/VCS	VCS Standard, Version 4.5	04.10.2023
49	VERRA/VCS	VCS Program Guide, Version 4.4	29.08.2023
50	VERRA/VCS	Validation completion deadline, Exemption mail	06.02.2023

APPENDIX 2: CLARIFICATION REQUESTS, CORRECTIVE ACTION REQUESTS AND FORWARD ACTION REQUESTS

CL from this validation

CL ID	01	Section no.	5.2	Date: 24/12/2022
Description of CL				
The calibration details of the monitoring equipment, namely electricity meters, flowmeters, laboratory devices, and weighbridge, have to be included in the relevant boxes.				
Project participant response				Date: 25/12/2022
The calibration details of the monitoring equipment have been given in the boxes of the relevant parameters.				
Documentation provided by project participant				
Revised VCS-PD version 2, Section 5.2				
DOE assessment				Date: 27/12/2022
The calibration information is provided as required in the REVISED VCS-PD. CL is closed.				

CL ID	02	Section no.	1.3	Date: 24/12/2022
Description of CL				
The eligibility of the project under the scope of the VCS program has to be justified, as per the VCS Program Standard V4.4, Section 2.1.1.				
Project participant response				Date: 25/12/2022
The scope of the VCS Program defined by the 5 items in Section 2.1.1 of the VCS Program Standard V4.4, is assessed and eligibility of the project is justified. Moreover, the project is also not among the excluded project activities given in Table 1 of the standard.				
Documentation provided by project participant				
Revised PD version 2, Section 1.3.				

DOE assessment	Date: 27/12/2022
The related section of the report is updated correctly by taking into account the scope defined under the applied standard. CL is closed.	

CL ID	03	Section no.	5.3	Date: 24/12/2022
Description of CL				
The PP is required to update the monitoring plan to include the monitoring details of all relevant parameters/variables as well as a clear diagram of organizational structure.				
Project participant response				Date: 25/12/2022
Monitoring plan has been updated to include detailed information of related parameters/data and a organizational structure chart has been included.				
Documentation provided by project participant				
Revised PD version 2, Section 5.3.				
DOE assessment				Date: 27/12/2022
The related section of the report is updated correctly by taking into account the monitoring actions required by the applied methodology. CL is closed.				

CL ID	04	Section no.	4.3	Date: 24/12/2022
Description of CL				
PP has to clarify why the leakage emissions were not taken into account.				
Project participant response				Date: 25/12/2022
Since the digestate produced during the project activity does not undergo composting and is distributed free of charge to the neighboring cropland and farm owners without any storage under anaerobic conditions, leakage emissions were not considered. Detailed explanation has been provided in the related section.				
Documentation provided by project participant				
Revised VCS-PD version 2, Section 4.3.				

DOE assessment	Date: 27/12/2022
The assessment team confirms the justification given in the REVISED VCS-PD. CL is closed.	

CL ID	05	Section no.	1.11	Date: 24/12/2022
Description of CL				
The PP is required to provide details of the digesters in the plant such as the operating conditions, capacities, etc.				
Project participant response				Date: 25/12/2022
The technical details of the digesters have been provided within a table in the related section.				
Documentation provided by project participant				
Revised PD version 2, Section 1.11.				
DOE assessment				Date: 27/12/2022
The related section of the report is updated to provide the necessary technical details of the digesters. CL is closed.				

CL ID	06	Section no.	3.3	Date: 24/12/2022
Description of CL				
The PP indicates in the table describing project boundary “This emission source is not accounted since the project does not consume electricity from the national grid” for the project scenario. However, the diagram under the table includes CO2 emissions from the grid in the project scenario. Please clarify.				
Project participant response				Date: 25/12/2022
The electricity generated from the gas engines in the plant is not only supplied to the national grid, but also utilized for auxiliary/internal consumption in the plant. Thus, electricity is not consumed from the grid, as it can be also observed from the electricity meter readings. The boundary diagram is corrected and CO2 emissions from the grid in the project scenario is removed from the figure.				
Documentation provided by project participant				
Revised PD version 2, Section 3.3.				

DOE assessment	Date: 27/12/2022
The assessment team checked the updates in the boundary diagram and finds the information consistent. CL is closed.	

CARs from this validation

CAR ID	01	Section no.	1.12 & Appendix-1	Date: 24/12/2022
Description of CAR				
The PP is required to clearly state names and locations of manure supplier farms in the relevant PD section. Further, since the applied methodology states that “if anaerobic lagoons are used in the baseline, their depths are at least 1 m”, the PP shall provide the details of lagoon depth for each identified supplier farm in the PD.				
Project participant response				Date: 25/12/2022
Names, locations, and lagoon depth information for each farm has been provided in Section 1.12 and Appendix-1.				
Documentation provided by project participant				
Revised PD version 2				
DOE assessment				Date: 27/12/2022
Required information on supplier farms have been included in relevant sections. CAR is closed.				

CAR ID	02	Section no.	1.12 & Appendix-1	Date: 09/12/2023
Description of CAR				
The PP is required to report $B_{0,LT}$ - Maximum methane producing potential of the volatile solid generated for animal type ‘LT’ parameter as an ex-post parameter in line with the applied Methodology				
Project participant response				Date: 10/12/2023
The status of related project has been changes as ex-post in line with the applied Methodology.				
Documentation provided by project participant				

Revised VCS-PD version 3	
DOE assessment	Date: 11/12/2023
$B_{0,LT}$ is reported correctly. CAR is closed.	

FAR from this validation

FAR ID	01	Section no.	1.12	Date: 09/12/2023
Description of FAR				
Any changes in farms included in the project activity at the time of registration shall be assessed during every subsequent verification and shall be reported as project deviation by the PP.				
Project participant response				Date: 10/12/2023
Any changes in farms included in the project activity at the time of registration will be reported as PD deviation in the upcoming verification periods.				
Documentation provided by project participant				
Revised VCS-PD Version 3.				
VVB assessment				Date: 11/12/2023
PP revised the VCS PD to reflect the requirements of this FAR.				

APPENDIX 3 ABBREVIATIONS:

Abbreviations	Full texts
BE	Baseline Emissions
CAR	Corrective Action Request
CDM	Clean Development Mechanism
CDM M&P	Modalities and Procedures CDM
CER(s)	Certified Emission Reduction(s)
CH ₄	Methane
CL	Clarification Request
CO ₂	Carbon dioxide
CO ₂ e	Carbon dioxide equivalent
DNA	Designated National Authority
EB	Executive Board
EF	Emission Factor
EPL	EcoLance Pvt. Ltd.
ER	Emission Reductions
FAR	Forward Action Request
GHG(s)	Greenhouse gas(es)
GWP	Global Warming Potential
IPCC	Intergovernmental Panel on Climate Change
LoA	Letter of Approval
MoV	Means of Verification
MWh	Mega Watt Hour
MR	Monitoring Report
NGO	Non-governmental Organization
ODA	Official Development Assistance
PDD	Project Design Document
PE	Project Emission
PP(s)	Project Participant(s)
Ref.	Document Reference

SS(s)	Sectoral Scope(s)
TA(s)	Technical Area(s)
UNFCCC	United Nations Framework Convention on Climate Change
VCS	Verified Carbon Standard
VCU	Verified Carbon Unit
VER	Voluntary Emission Reductions
VVB	Validation and verification body
VVS	Validation and Verification Standard

APPENDIX 4: TEAM COMPETENCY CERTIFICATES



Shailendra Kewat

has been qualified for below schemes & Sectoral Scopes as per the requirements of ECOLANCE Private Limited to participate in validation/verification audits:

Standard	ISO 14064-1	ISO-14064-2	VCS
Qualified	☒	☒	☒

Role	Validator	Verifier	TL	Technical Reviewer	Technical Expert
	☒	☒	☒	☐	☒

Scopes Granted:

- Power Generation and Electric Power Transactions (Fossil fuel thermal/ Renewable energy non thermal)
- Energy Industries (Renewable/Non renewable sources)
- Energy Demand
- Waste handling and disposal

Version	DATE OF APPROVAL	VALIDITY
1	12 December 2022	11 December 2023

Authorized Signatory



Rekha Menon
Director- Technical

EPL/FR/5

ECOLANCE PRIVATE LIMITED

Version 1

Issued: 27-10-2021



Champok Buragohain

has been qualified for below schemes & Sectoral Scopes as per the requirements of ECOLANCE Private Limited to participate in validation/verification audits:

Standard	ISO 14064-1	ISO-14064-2	VCS
Qualified	☒	☒	☒

Role	Validator	Verifier	TL	Technical Reviewer	Technical Expert
	☒	☒	☒	☒	☒

Scopes Granted:

- Power Generation and Electric Power Transactions (Fossil fuel thermal/ Renewable energy non thermal)
- Energy Industries (Renewable/Non renewable sources)
- Energy Demand
- Waste handling and disposal

Version	DATE OF APPROVAL	VALIDITY
1	10 November 2021	09 November 2022

Authorized Signatory



Rekha Menon
Director- Technical

EPL/FR/5

ECOLANCE PRIVATE LIMITED

Version 1

Issued: 27-10-2021